FocalFlow: Facilitating Decision-Making in Multi-Step Mobile Tasks for Blind and Low Vision People

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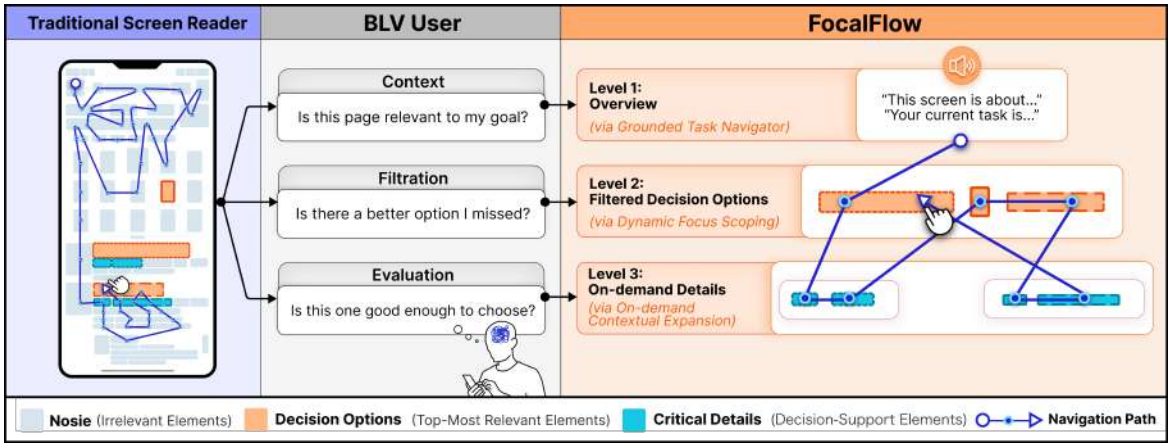


Fig. 1. Conceptual comparison of decision-making workflows. (Left) Using traditional screen readers, BLV users often engage in random exploration, gathering incomplete information that forces them to make compromised decisions. (Center) Three key questions, derived from our formative study, illustrate the confusion BLV users face. (Right) FocalFlow facilitates informed decision-making by organizing interface information into three levels of granularity (Levels 1–3), explicitly designed to answer these questions. Corresponding colored elements in the left and right panels represent the same content objects.

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Multi-step decision-making tasks, such as booking a hospital appointment, on mobile devices present a two-fold challenge for blind and low vision (BLV) users: they might struggle to gather relevant information and make informed decisions, as screen readers present task-relevant elements sequentially alongside irrelevant ones. We present FocalFlow, an LLM-powered system that (1) decomposes tasks into actionable subtasks and provides an error recovery process, (2) dynamically filters out task-irrelevant information while preserving the original interface layout, and (3) utilizes progressive disclosure to present critical information first, with decision-support details available on demand. Our evaluation with twelve BLV participants demonstrated that FocalFlow enables them to engage in more deliberate decision-making. We then discuss design implications for future non-visual systems to balance automated information curation with user agency.

CCS Concepts: • **Human-centered computing** → **Accessibility systems and tools**; *Empirical studies in accessibility*; *Mobile devices*.

Additional Key Words and Phrases: Accessibility, Mobile Devices, Large Language Models.

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1 INTRODUCTION

Modern mobile applications have evolved beyond simple actions into complex multi-step task flows comprising multiple decision points, such as booking a flight or ordering groceries [15, 23, 55]. For instance, scheduling a hospital appointment requires a user to select a department, choose a doctor based on specializations and availability, and confirm a specific date and time. By leveraging spatial layouts, visual groupings, and hierarchical structures, graphical user interfaces (GUIs) allow sighted users to scan options, compare alternatives simultaneously, and efficiently gather pertinent information while filtering out irrelevant data [43, 83]. However, for blind and low vision (BLV) users who rely on screen readers, this rich visual information is linearized into a sequential stream of audio feedback. In contrast to web interfaces, which typically present dense information on a single scrollable plane, mobile applications are characterized by significant hierarchical depth. Due to limited screen real estate, cohesive decision-making processes, such as booking an appointment, are fragmented into multi-step sequences distributed across distinct screens. This process does more than just impede efficiency [13, 14, 42, 51, 103], it fundamentally obstructs the cognitive processes of informed decision-making. When BLV users try to make a hospital appointment, they must first overcome a barrage of irrelevant information just to locate potential options. To compare options, they are then forced to hold the details of one doctor in their working memory while navigating through more noise to find the next. Consequently, making a simple choice is a significant cognitive challenge for BLV users. In this work, we focus specifically on goal-directed, multi-step decision-making tasks—settings in which the user holds a relatively well-defined goal prior to engaging with the interface.

To bridge this accessibility gap, both academia and industry have developed solutions aimed at improving navigation [119]. Modern operating systems, for instance, integrate structured navigation tools such as the "Rotor" in Apple's VoiceOver [8] and "Reading Controls" in Android's TalkBack [33]. These allow users to jump between specific element types like headings or links. However, while these features improve efficiency for navigation, their filtering mechanism is context-agnostic, often misaligned with the user's goals, and its effectiveness is dependent on developers' adherence to accessibility guidelines (such as WCAG) [21, 27, 86, 87, 111, 112, 117]. Consequently, BLV users are often forced to revert to an exhaustive, sequential navigation method [78], sifting through numerous task-irrelevant elements, which obstructs the core decision-making process, and sometimes result in negative outcomes, such as booking a suboptimal appointment or abandoning the task altogether [13, 78].

Prior research has established that removing task-irrelevant elements from web interfaces can reduce task completion time [29, 72], minimize user distraction, and show potential in facilitating the decision-making process. However, the unique interaction characteristics of mobile devices present new challenges and opportunities that

have yet to be fully explored. Many BLV users navigate via "*touch exploration*," relying on a stable mental map of the screen's layout for efficiency (e.g., knowing the back button is on the top-left corner) [78]. Directly removing or hiding interface elements disrupts the layout, undermining BLV users' established navigation strategies and impairing usability, particularly in mobile scenarios. Furthermore, BLV users still struggle to grasp the purpose of a new screen and the action required to proceed, especially when engaged in multi-step tasks [14].

Addressing this challenge requires a system that can interpret task context, restructure information, and present it to BLV users in a manner that facilitates decision-making. The emergence of Large Language Models (LLMs), with their powerful capabilities in comprehending dynamic interfaces and complex instructions [28, 47, 56, 60, 65, 96], provides a robust technological foundation for such a system. A prominent research direction involves leveraging LLMs to automate tasks by deploying agents to operate on the user's behalf [41, 67, 70]. However, fully automated processes often prove insufficient for addressing the specific needs of BLV users, highlighting the necessity for highly customizable interaction methods [57, 90].

To better understand how LLMs can support BLV users to make more deliberate and informed decisions, we conducted a formative study in which 12 BLV users completed multi-step mobile tasks using screen readers. Our research identified three primary challenges faced by BLV users in multi-step tasks: (1) difficulty in maintaining orientation within a task flow and recovering from errors, (2) cognitive overload from navigating irrelevant elements to find the target element, and (3) the risk of missing decision-critical details.

To address these challenges, we propose FocalFlow, an LLM-driven system that enhances the standard screen reader experience. It introduces three novel features: (1) a ***Grounded Task Navigator*** that decomposes a task into manageable subtasks by analyzing the current interface and overall task objective, and offering a seamless error recovery process; (2) ***Dynamic Focus Scoping***, a feature that dynamically filters out task-irrelevant information while preserving the original interface layout; and (3) ***On-demand Contextual Expansion***, which allows BLV users to request detailed decision-support information on demand without being overwhelmed by default verbosity.

To evaluate FocalFlow, we conducted a comparative user study involving 12 BLV participants who performed a series of complex multi-step tasks within real mobile applications. The results demonstrate that FocalFlow enabled BLV users to make more satisfactory and well-informed decisions with less cognitive load. Furthermore, all participants indicated that FocalFlow provided them with more opportunities to make deliberate decisions throughout the workflow and showed a strong desire to use FocalFlow in the future. We conclude by discussing key implications and addressing generalizability concerns.

2 RELATED WORK

2.1 Limitations of Screen Readers on Mobile Interfaces

Screen readers are assistive technologies that convert visual information from a user interface into speech, enabling BLV users to interact with digital devices [25]. However, graphical user interfaces (GUIs) are inherently visual, relying on a rich set of cues such as spatial layout, element grouping, and a visual hierarchy established through attributes like color and size. This fundamental mismatch between visual GUIs design and non-visual access methods results in a significant loss of navigational efficiency and a heightened cognitive burden[14, 22, 45, 55, 89, 93, 121].

While structured navigation features allow jumping between specific elements (e.g., headings) [112], their efficacy is frequently undermined by prevalent deficiencies in accessible development practices[48, 78]. Such deficiencies include the improper use of semantic HTML on websites and the failure of many mobile applications to meet platform-specific accessibility standards (e.g., not labeling interactive elements correctly). Consequently, these developmental shortcomings often compel BLV users to revert to a less efficient, sequential navigation method[84].

Although screen readers provide essential access to information, they also impose a significant cognitive load [26, 89]. The responsibility for filtering, structuring, and synthesizing the linear stream of auditory information falls entirely upon the BLV users. Consequently, when performing complex, multi-step mobile tasks, this burden requires users to expend considerable time and energy evaluating the utility of each element, thereby diverting focus from the core decision-making process [50].

2.2 Barriers to Non-Visual Decision-Making in Multi-Step Tasks

Humans are subject to three critical constraints when making decisions: limited information, limited cognitive processing capacity, and limited time [94, 95]. In the context of multi-step mobile tasks, these three constraints are exacerbated for BLV users due to the mismatch between parallel layout of visual information in GUIs and sequential nature of information delivery by screen readers[14, 22, 89, 93]. BLV users have difficulty in forming a comprehensive mental model of the information (limited information) [48, 102] and must expend considerable cognitive effort to memorize and mentally organize the fragmented auditory content they receive (limited cognitive processing) [5, 89, 98], leading to a more time-consuming and arduous process (limited time) [13, 14]. Consequently, this mismatch critically undermines the foundational phases of any structured decision-making process [24, 46, 68]: Problem Identification (i.e., understanding the task's scope and goal from the interface) and Information Collection (i.e., systematically gathering relevant data to evaluate options).

2.2.1 Problem Identification. Researchers have proposed various methods to help BLV users better comprehend the information space of a task so that they can identify the problem and clarify what actions are possible. Li et al. [61] proposed TangibleGrid, a tangible shape-changing interface that provides real-time physical feedback to help BLV users comprehend and design web page layouts. Khurana et al. [52] developed Spatial Region Interaction Techniques (SPRITEs) that leverages the keyboard surface to enable nonvisual navigation of two-dimensional structures. However, these methods still require BLV users to define the problem themselves based on the information they have gathered, which still consumes cognitive resources. Our approach utilizes task scaffolding to automatically decompose a task into multi-step subtasks, proactively guiding the user through each step.

2.2.2 Information Collection. Prior work has focused on two approaches to help BLV users efficiently collect information: content modification and user customization. Content modification includes restructuring [93, 118, 119], summarization [7, 12], dynamic text segmentation [63], and adding semantic markup [13, 35] or labels [10, 110] or physical tactile markers [31]. User customization involves enabling features like personal overlays [104] or user-inserted navigation points for skimming [12]. Despite the improved efficiency of these methods, BLV users still need to navigate a large amount of information irrelevant to their goals [7, 9, 12, 80, 81]. Previous studies have demonstrated that removing task-irrelevant elements can help BLV users reduce task completion time[29, 42, 72]. However, most of this research has focused on web scenarios, leaving the unique challenges of mobile navigation still unaddressed. In contrast, mobile interaction relies heavily on "*Touch Exploration*," where users build a mental model based on the absolute physical location of elements. On mobile devices, directly removing or hiding elements causes layout reflow, which disrupts spatial consistency. This distinction necessitates that BLV users rely on mental models of an interface's spatial layout to navigate efficiently [49, 54, 64] (e.g., remembering the typical location of a back button). Yet, the practice of directly removing elements often alters the page's spatial structure, which undermines the mental models[37, 75]. To address this gap, our method automatically filters out irrelevant information without changing the original interface layout, thereby enhancing navigation efficiency on mobile devices while upholding the integrity of the mental models that BLV users depend on.

2.3 AI-Supported Task Operation in Mobile Accessibility

While early work relied on human-powered crowdsourced assistance [85], recent advances in agents have enabled AI-supported task operation. Prior work on AI-driven support for BLV users has largely focused on two areas: enhancing access to visual content and developing more flexible interaction paradigms. Efforts to improve access to visual content include systems for real-time scene, object, and text description [30, 71], as well as tools for interactively exploring or summarizing image and video content [54, 74, 107]. In parallel, other research has focused on creating more intuitive and personalized interaction methods, such as conversational assistants [76, 79, 82], and teachable AI systems that allow users to customize behavior [114]. However, the majority of existing research focuses on single-page or single-session tasks, while the more common requirement for users to complete multi-step tasks in real-world applications remains comparatively underexplored. Due to the limited screen space on mobile devices, designers often organize information in layers to keep the interface clean [18, 58]. This approach requires BLV users to tap or swipe multiple times to navigate content, which increases the number of steps needed to complete a task.

Large Language Models (LLMs) have driven a major shift in task automation, owing to their capacity to comprehend dynamic interfaces and complex instructions [44, 47, 56, 60, 65, 96]. This has enabled a new generation of AI agents that serve as intermediaries between users and digital environments, interpreting user intent to perform complex tasks like navigation, form filling, and multi-step operations across various platforms [41, 67, 70]. These agents leverage LLMs to infer UI state, detect actionable elements, and dynamically generate or simulate user interactions [106, 116]. Mobile Agent [108, 109], AppAgent [120] and CogAgent [39] exemplify this approach by combining visual cues, human demonstrations, and trial-and-error learning to automate complex app-based workflows. This paradigm has also been extended to enhance accessibility: HintDroid and VisionTasker adapt these models to assist BLV users in navigating applications or switching contexts without manual setup [66, 97].

Prior research on task automation using LLMs has predominantly focused on fully autonomous, end-to-end methods. However, this approach often overlooks the critical decision points inherent in real-world, multi-step mobile tasks, consequently diminishing user agency. To address this limitation, FocalFlow assists BLV users by identifying key decision points on each interface. The system then curates and restructures relevant information while leaving other critical decision-making steps to the users, thereby preserving their control and autonomy.

3 FORMATIVE STUDY

To better understand how LLMs can support BLV users to make more deliberate and informed decisions, we conducted a formative study with twelve BLV participants who use screen readers frequently in their daily activities. Through interviews and observation, we aimed to answer: what are the primary obstacles that hinder BLV users from efficiently gathering and synthesizing the information needed to make an informed decision?

3.1 Methods

3.1.1 Participants. We recruited 12 BLV participants (five males, seven females) who use screen reader in their daily activities. Their demographic information is shown in Table 4. These participants were recruited from the local community, with ages ranging from 23 to 43 ($Mean$ = 30.3, SD = 5.8). Nine participants were totally blind and three participants were low vision. The reported age of onset of vision impairment varied from 1 to 9 years ($Mean$ = 3.4, SD = 3.0). The participants were highly experienced screen reader participants, with a mean experience of 9.9 years (SD = 2.3), ranging from 3 to 12 years. Each participant received a compensation of approximately 14 USD. The study was approved by the Institutional Review Board (IRB) of our institution.

3.1.2 Procedure. The study consisted of two successive phases: a 20-minute observation followed by a 45-minute interview, with a 5-minute break in between. And all participants filled up informed consent form before the study.

Phase 1: Observation. We instructed BLV participants to employ their personal smartphones and everyday screen readers to navigate a local online travel agency (OTA) application and select a specific destination and purchase a train ticket. They were encouraged to think aloud about any confusion during the process. The entire session was recorded for subsequent analysis.

Phase 2: Semi-structured Interview. a post-task semi-structured interview covered participants' performance, observed behaviors, and broader information-management practices on mobile apps

3.1.3 Data Analysis. We transcribed, annotated all audio and screen recordings from the study sessions and employed an open-coding[99] approach grounded in the data. Two researchers independently open-coded transcripts and screen-recording behaviors, then iteratively clustered codes into themes through joint discussion until consensus.

3.2 Findings

Our analysis indicated that participants' difficulties extended beyond navigation to the full multi-step decision-making process—from maintaining task context, through identifying relevant information, to evaluating options with sufficient context. Fig. 9 illustrates how these breakdowns unfolded across a representative task flow.

C1: Difficulty in maintaining orientation within a task flow and recovering from errors. The first challenge was the difficulty participants experienced in reorienting themselves after a change in the interface context. BLV participants often became disoriented following screen transitions. We observed that when a specific interaction (e.g., selecting a date) triggered a view change, most participants (N = 8) lost their mental model of the page structure. This phenomenon was explicitly described by P2: "*I think I selected the date, but the calendar view is gone. Am I back on the main booking screen?*" Consequently, this process of rebuilding a mental model of their location within the application was a significant drain on time and cognitive resources.

C2: Cognitive overload from navigating irrelevant elements to find the target element. Second, all participants ($N = 12$) reported that distinguishing relevant from irrelevant content imposed a high cognitive load. This burden intensified when participants accidentally navigated to erroneous contexts, such as ads, without realizing they had deviated from the main task. Lacking clear signifiers, they often continued scanning incorrect pages for non-existent targets. P8 illustrated this frustration:*"I kept looking because I didn't know if the button wasn't there or if I just hadn't reached it yet. Sometimes, I really don't know if the page has what I need."* In such cases, decision-making devolved into a process of pure luck and trial-and-error.

C3: The risk of missing decision-critical details. Third, critical decision parameters (e.g., baggage allowance) are often fragmented and distant from confirmation actions, creating "blind spots" in the decision-making process. The high volume of interface elements hinders associating descriptive details with corresponding options. Consequently, participants frequently committed to actions without full context. When researchers asked detailed follow-up questions about the tickets they had selected, half of the participants (N=6) realized for the first time that they had missed critical constraints. Their common reaction was one of surprise: *"That doesn't seem to be what I wanted. I think the screen reader didn't read that part out just now."* This suggests that without a mechanism to synthesize distributed information, BLV users may face challenges in making fully informed decisions.

Overall, our formative study contributes a complementary observation: in goal-directed multi-step workflows, these barriers compound across screen transitions rather than occurring in isolation — disorientation forces mental-model rebuilding on every screen, the resulting cognitive load impedes filtering, and fragmented attributes mean users may only realize what they missed after committing. This coupling motivates a workflow-aware design that treats orientation, filtering, and on-demand context as interdependent. To address these challenges, we designed ***FocalFlow***.

4 FOCALFLOW SYSTEM

To address the challenges identified in the formative study, we propose FocalFlow, which introduces three core features:

(1) ***Grounded Task Navigator***: To address the disorientation in the mobile scenarios, where a single task is split across multiple sequential screens, FocalFlow offers a high-level summary of the current screen and suggests a primary subtask. This provides essential interface context, thereby preventing users from losing track of their progress within the workflow.

(2) ***Dynamic Focus Scoping***: To reduce clutter and improve efficiency, FocalFlow intelligently restricts the screen reader's focus to a limited set of UI elements that are most relevant to the current subtask, while strictly preserving the original visual coordinates. This design respects the "*Touch Exploration*" paradigm, ensuring that users can still rely on their spatial memory to locate buttons without encountering unexpected layout shifts.

(3) ***On-demand Contextual Expansion***: To ensure user control, FocalFlow allows users to request context or adjust the focus scope using a simple gesture (triple-tap with two fingers) followed by a natural language command.

In the following sections, we first present a representative user walk-through in Section 4.1. Next, we describe the overall system architecture and implementation in Section 4.2, followed by our task-oriented interface modeling of UI elements in Section 4.3. Finally, we elaborate on the implementation details of each feature in Section 4.4-4.6.

4.1 User Walk-through

To illustrate the system's workflow and features, Fig. 2 shows how FocalFlow supports a user to make a deliberate and informed decision.

4.2 System Architecture and Implementation

FocalFlow realizes its three features through an Android accessibility service and a laptop-hosted agent-control backend, which invokes external LLM and speech APIs in the evaluated prototype. Figure 3 summarizes these components, their deployment locations, and the primary data exchanged among them.

4.2.1 Agents Supporting the Three Features. The system comprises three collaborative agents, each assigned a distinct role:

- ***Planning Agent:*** Responsible for global task management and context grounding. It performs four operations: (i) judges whether the previous *Current Subtask* has been accomplished and, if so, appends it to *Completed Subtasks*; (ii) determines whether the *Overall Task Objective* has been satisfied, signaling termination; (iii) generates a high-level *Screen Overview* of the new screen; and (iv) generates the next *Current Subtask*. Operations (iii) and (iv) drive the user-facing *Grounded Task Navigator* (Section 4.4).
- ***Decision Agent:*** Acts as the primary filter for interface clutter. it performs two operations: (i) identifies the set of Semantic Component Blocks (SCBs) on the current screen that are relevant to the *Current Subtask*; and (ii) classifies on-screen elements into three *Task-Oriented Elements*, and refines the initial *VAT* into an Updated *VAT*. The Updated VAT is then applied by the Android client to drive the user-facing *Dynamic Focus Scoping* (Section 4.5).
- ***Talk Agent:*** Manages on-demand natural language interactions. It is invoked only when the user performs a triple-tap-with-two-fingers gesture, and performs two operations: (i) interprets the transcribed *User's Instruction* against the current screen, identifying what information the user is requesting; and (ii) further refines the Updated *VAT* to expose the requested *Decision-Support Elements* as additional virtual focus nodes.

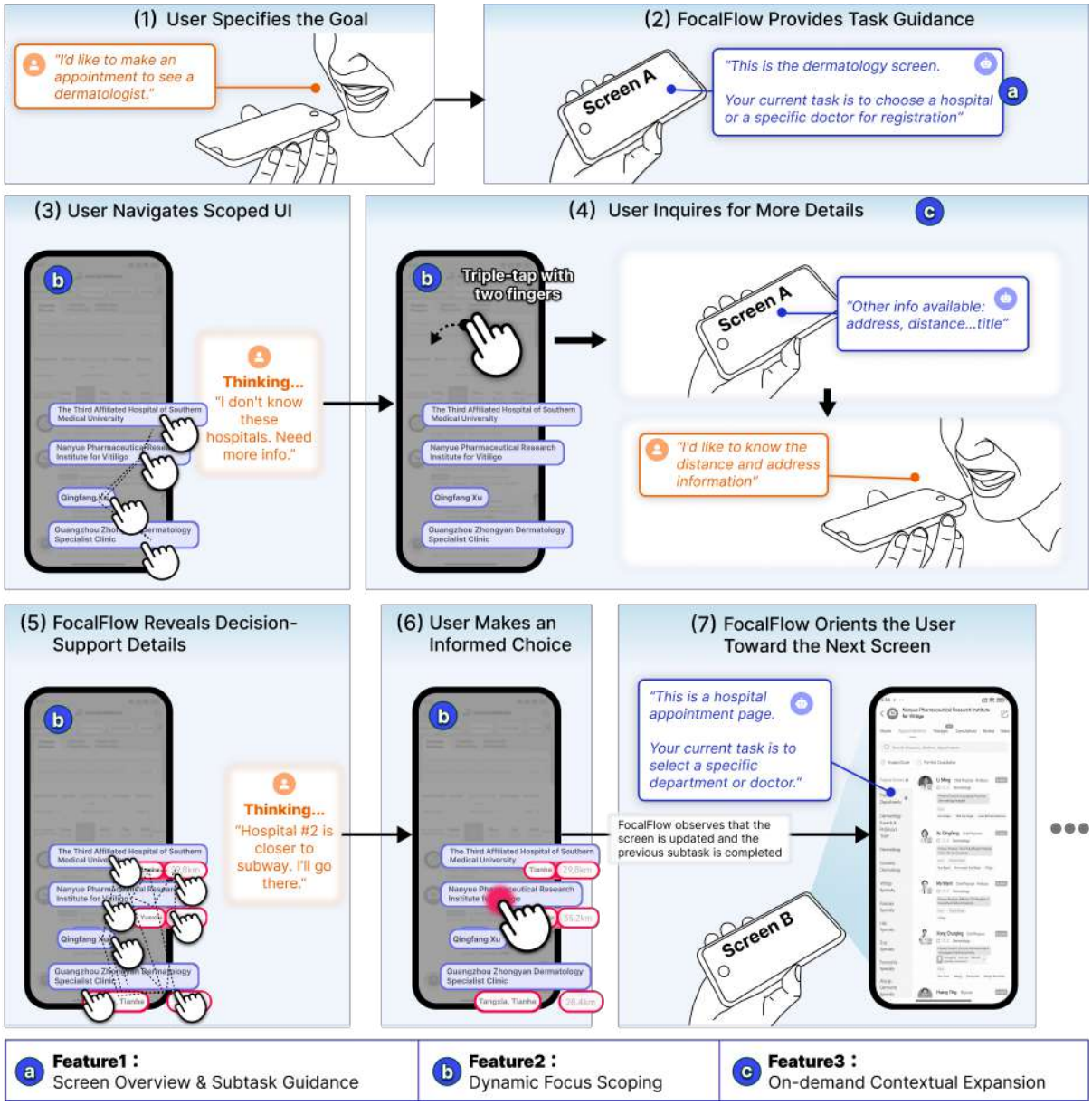


Fig. 2. An end-to-end walk-through of a user interacting with FocalFlow during a multi-step task (e.g., booking an appointment). (1-2) The user initiates the task via voice, and FocalFlow orients them to the current interface (Screen A) with a clear overview and subtask. (3) The user navigates through a dynamically scoped active area containing only top-most relevant options. (4-5) Needing more details, the user invokes on-demand contextual inquiry via a two-finger triple-tap gesture, expanding the focus to include decision-support information (e.g., distance). (6-7) Upon making an informed decision and advancing the interface, FocalFlow continuously tracks the workflow, seamlessly orienting the user to the new context (Screen B) for the next step of the multi-step task.

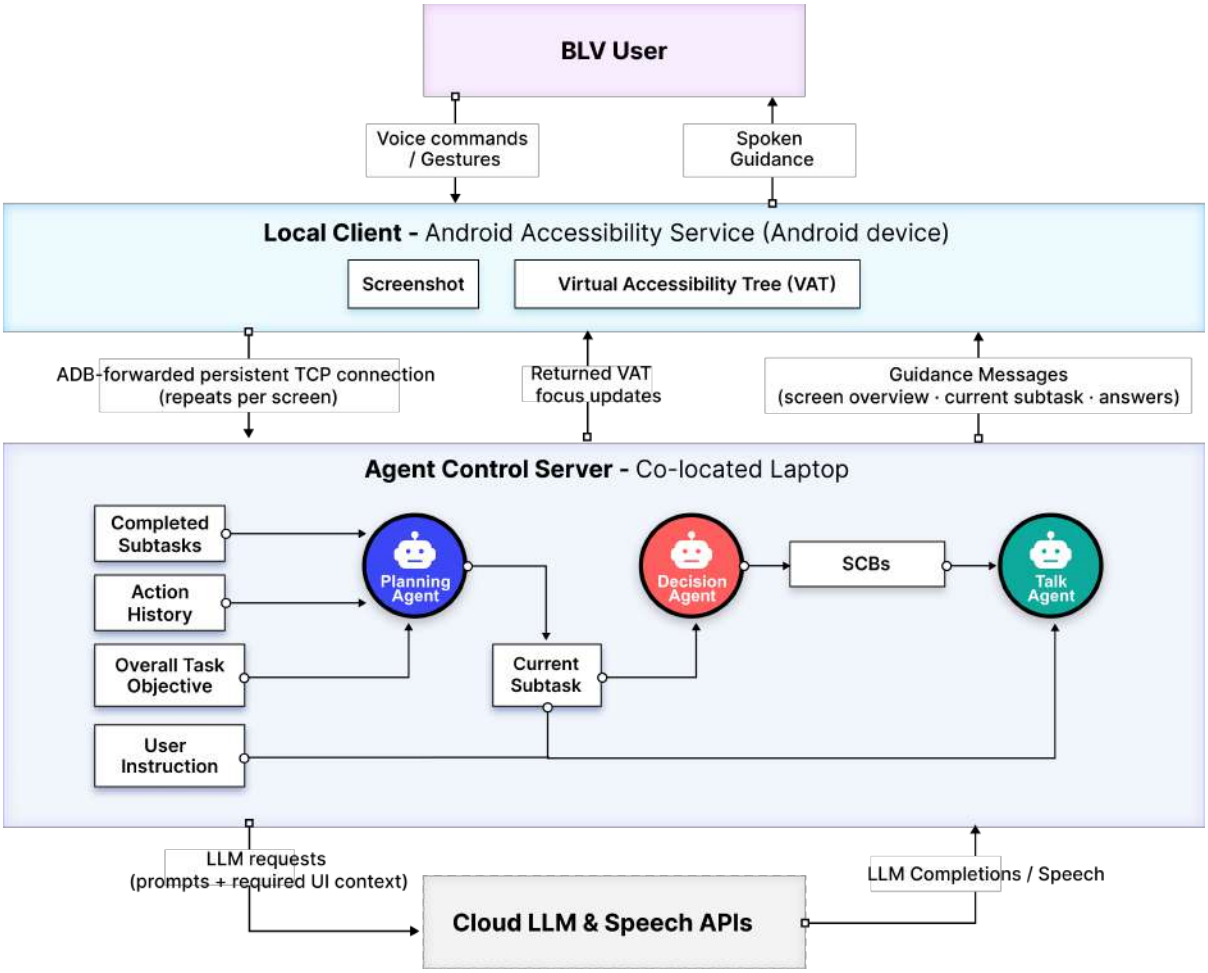


Fig. 3. FocalFlow system architecture. On the Android device, the local client captures the current UI state, receives user task or query input, applies returned VAT focus updates, and delivers spoken guidance. The laptop-hosted agent-control server maintains task state and coordinates the Planning, Decision, and on-demand Talk agents. It invokes an external cloud LLM API and the system's speech-transcription and synthesis components. The phone and laptop communicate through an ADB-forwarded TCP connection.

The refined VAT is applied by the Android client to drive the user-facing *On-demand Contextual Expansion* (Section 4.6).

4.2.2 End-to-end Operation Flow. A complete FocalFlow session unfolds in three phases: initialization, a recurring per-screen cycle, and termination.

Initialization. A session begins when the user states a high-level intent. FocalFlow then transcribes the utterance into the *Overall Task Objective*, which persists unchanged for the remainder of the session. *Action History* and *Completed Subtasks* are initialized as empty lists. The Android client immediately captures the current screen and the system enters the per-screen cycle.

Per-screen cycle. On each activity transition, the Android client extracts the raw accessibility tree together with a screenshot, and constructs an initial *VAT* as a one-to-one overlay of the raw tree. The *Planning Agent* is invoked first. Its outputs (*Screen Overview* and *Current Subtask*) are delivered to the user via TTS and forwarded to the *Decision Agent*.

The *Decision Agent* then identifies relevant SCBs and refines the VAT, which the client applies by overriding focus traversal to step only through virtual focus nodes. Any user action resulting in a screen transition (e.g., a double-tap) is appended to *Action History*. At any point, a triple-tap-with-two-fingers gesture invokes the *Talk Agent*, which interprets the *User's Instruction* verbally and further refines the Updated *VAT* to expose the requested decision-support elements. The cycle repeats whenever the Android Accessibility Service detects a structural UI change (e.g., via window state events).

Termination. The session ends when the Planning Agent determines that no further subtask is required to satisfy the *Overall Task Objective*, at which point a completion confirmation is announced and the system returns to idle.

4.2.3 Implementation. The proposed system operates on a client-server architecture that facilitates real-time, bidirectional interaction between an Android mobile device and a host server. This connection is maintained via a persistent TCP socket established through Android Debug Bridge (ADB) port forwarding. On the client side, the module is implemented as an Android accessibility service derived from Google's TalkBack framework and was validated on a Redmi Note 12T Pro. This component leverages the standard AccessibilityNodeInfo class to extract the Accessibility Tree from the active screen, capturing essential structural and semantic data (e.g., node hierarchy, text, and content descriptions). It also applies the VAT generated by the agent layer by overriding the screen reader's focus traversal. Conversely, the backend processing is executed within a Python environment on a MacBook Pro equipped with an M3 Pro chip. This host supports a multi-agent collaborative framework powered by the GPT-4o model [3] (configured with a temperature of 0) to interpret raw UI data and user instructions. Additionally, the system integrates gpt-4o-mini-tts for text-to-speech (TTS) synthesis and Whisper for speech-to-text (STT) transcription to handle multimodal user interactions.

4.3 Task-Oriented Interface Modeling

To enable task-oriented interface adaptation, we model the interface along two complementary dimensions: a taxonomy that classifies every element by its role in the Current Subtask (Section 4.3.1), and a regional grouping of task-useful elements into semantically coherent units (Section 4.3.2). The Virtual Accessibility Tree (Section 4.3.3) then operationalizes both to redefine what the screen reader can reach.

4.3.1 Task-Oriented Element. As shown in Fig. 4, we categorized on-screen elements into types based on their utility to the user's current subtask:

- **Top-most Relevant Elements:** These are the primary interactive components strictly required to execute the current subtask. They represent the minimum set of interactive nodes that the user must access to proceed.
- **Decision-Support Elements:** This category encompasses auxiliary information that facilitates the evaluation of the Top-most Relevant Elements. While these elements are not strictly required to trigger the next action, they provide critical semantic context that informs the user's decision-making process
- **Irrelevant Elements:** Any component that contributes neither to the execution of the current subtask nor to the immediate decision-making process is classified as irrelevant. This includes status icons, advertisements, or content related to unrelated workflows.

4.3.2 Semantic Component Block (SCB). While the element-level classification above identifies which elements are useful, it does not capture which useful elements belong together. On a screen listing multiple hospitals,

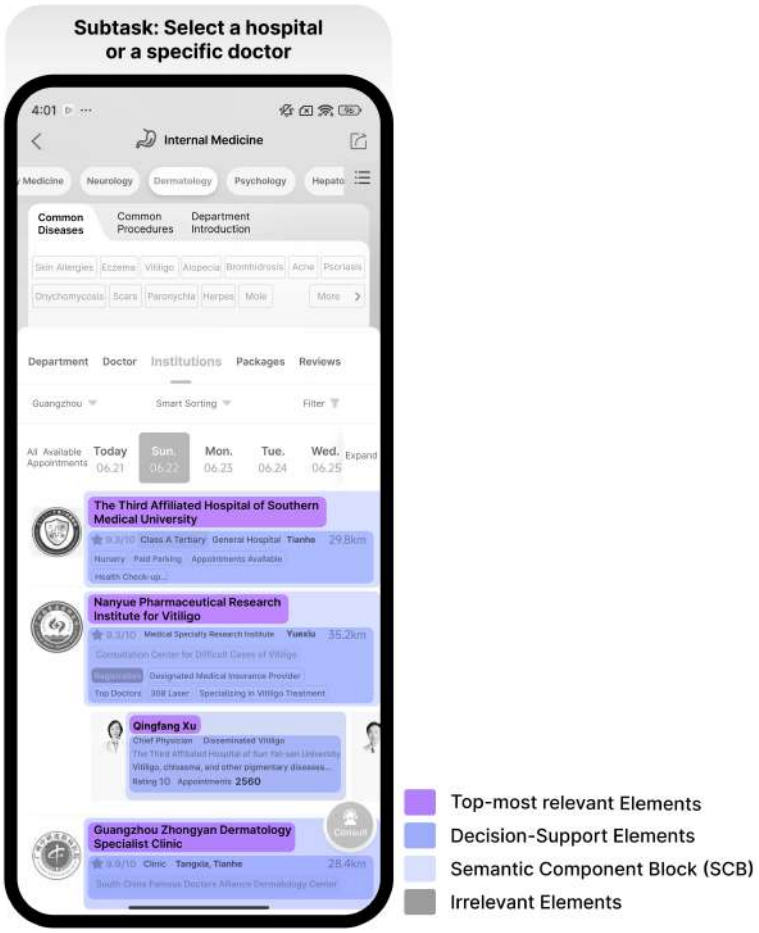


Fig. 4. Illustration of our task-oriented UI element classification. Each SCB contains two types of elements (Top-most relevant elements and Decision-support elements), indicated by different colors. Irrelevant elements are located outside the SCB.

for instance, the user should still know which rating belongs to which hospital. To preserve such intra-group semantic associations, we introduce the SCB.

We define an SCB as a self-contained, visually grouped region of the interface (typically a card, list item, or panel) whose contents collectively support a single decision point in the user's workflow. Each SCB encapsulates the Top-most Relevant and Decision-Support Elements belonging to that decision point and Irrelevant Elements lie outside any SCB, as shown in Fig. 4.

4.3.3 Virtual Accessibility Tree (VAT). To operationalize these concepts, we introduce the VAT, which serves as the mechanism for redefining the screen reader's active area. The VAT is a dynamic overlay applied to the original physical accessibility tree. It is constructed based on two mechanisms:

Virtual Focus Nodes. We introduce a semantic attribute, *isVirtualFocusNode*, to distinguish task-relevant elements from noise. The VAT reconfigures the screen reader's traversal path to strictly follow nodes marked with this attribute.

Interaction Mapping. While the VAT restricts the navigation path to reduce cognitive load, it preserves a structural mapping to the underlying physical nodes. This ensures that user intents targeting a virtual node can be accurately projected back to the native accessibility tree for interaction execution, maintaining functional fidelity with the original app.

4.4 Feature 1: *Grounded Task Navigator*

To orient BLV users, FocalFlow employs a *Planning Agent* that generates two-part voice messages, providing BLV users with both screen overview and subtask guidance.

4.4.1 Screen Overview. The first component of the agent's output provides a concise summary of the current screen's function to quickly ground the user. To achieve this, we employ a few-shot prompting strategy to guide LLM toward generating a high-level functional description[19], intentionally omitting any mention of specific UI element names (e.g., "*This screen is for selecting a flight*").

4.4.2 Subtask Guidance. ***Operational Definition.*** In FocalFlow, a subtask is a runtime artifact generated by the Planning Agent: a single, atomic user-intent unit that represents the next decision the user should make on the current screen. It is regenerated whenever the screen state changes.

Decomposition Mechanism. We adopt a just-in-time, per-screen decomposition rather than producing a full subtask plan upfront. On each screen transition, the Planning Agent emits exactly one next subtask. This design is motivated by two factors: (1) full upfront plans are brittle when downstream screens contain unanticipated states (e.g., an ad that pops up out of nowhere); and (2) screen transitions are precisely where BLV users lose orientation (C1 in the formative study), so aligning subtask boundaries with screen boundaries gives users a stable anchor at each reorientation point.

Mitigating LLM Variability. To mitigate the inherent variability of LLM outputs, we design a subtask generation scheme that constrains generation along three axes: (a) History-grounded: the agent conditions on completed subtasks, lowering the chance of redundant or out-of-order outputs. (b) Layout-agnostic: subtasks are phrased as abstract intents (e.g., "select a hospital") rather than UI operations, making outputs less sensitive to surface UI variation. (c) Single-intent: each subtask encodes one decision, narrowing the granularity range of possible outputs [69].

4.5 Feature 2: *Dynamic Focus Scoping*

Once BLV users comprehend their current progress and the primary task, then they need to navigate the interface to locate the elements necessary to proceed. To address this, FocalFlow streamlines user interaction by redefining the screen reader's active area. This dynamic adjustment filters out UI elements on the current screen that are irrelevant to the subtask. Consequently, all elements reachable by the screen reader's focus are directly related to the current objective. This redefinition of the active area is orchestrated by a *Decision Agent*. The main functions of the *Decision Agent* are described below.

4.5.1 Identify SCB. The critical misalignment of visual data with generated text due to LLM hallucination is a significant challenge in multimodal and long-context applications[40]. To mitigate this issue, we utilize a filtered representation of the UI's view hierarchy. Specifically, the view hierarchy of the current screen is pruned to retain only the *text* and *contentDescription* attributes, alongside a unique identifier for each element. This distilled structure is then treated as the ground truth for the UI's semantic content.

However, the accessibility node tree in Android does not always maintain a one-to-one correspondence with the view hierarchy[1]. Consequently, parent-child nodes situated within a single SCB are at times mistakenly presented as sibling nodes. Such an erroneous assessment of the hierarchical structure can lead to the mispositioning of the SCB, resulting in the loss of valuable information. To overcome this limitation, we incorporate a screenshot into the prompt, providing *the Decision Agent* with the necessary visual context to discern the layout, grouping, and relative importance of elements. *The Decision Agent* then outputs a curated list of unique identifiers, each corresponding to a SCB.

4.5.2 Create VAT. To ensure the VAT is accurate and actionable, the *Decision Agent* performs a secondary refinement by classifying task-related attributes within each SCB. This granular filtering identifies the optimal action point and is governed by two critical requirements. First, the chosen action point must be clickable, ensuring it is an interactable element. Second, it must provide clear action context. To this end, if a selected action point contains generic or non-descriptive text, *the Decision Agent* is required to augment it by including other key textual elements from the same SCB, thereby clarifying the action's specific target.

4.6 Feature 3: *On-demand Contextual Expansion*

While the *Dynamic Focus Scoping* function effectively streamlines interaction, its minimalist nature may occasionally omit contextual information that BLV users deem necessary for confident decision-making. To address this challenge, FocalFlow provides an on-demand, interactive mechanism for unveiling additional UI elements when the user feels the initial active area is insufficient. This function is triggered by a dedicated gesture *Triple-tap with two fingers*, shifting the FocalFlow into an interactive mode to selectively expand the focus.

Upon activation, the *Talk Agent* is invoked. It is designed to perform two main functions: First, the *Talk Agent* proactively informs the BLV user about the categories of elements present on the SCBs. Based on the user's verbal feedback, it then adds the requested elements to the active area and dynamically updates the VAT. Second, it provides off-screen contextual awareness. If a user asks about information beyond the scope of the current SCB, *the Talk Agent* leverages the full screenshot to resolve their query. For example, when the focused area shows three hospitals but additional hospitals are visible elsewhere on the screen, the *Talk Agent* can inform the user that more options are available by scrolling. This helps the BLV users build a more accurate mental model of the entire interface.

5 TECHNICAL EVALUATION

To assess FocalFlow's LLM pipeline, we evaluated all three agents on AndroidControl [62], measuring accuracy across subtask inference, SCB identification, element classification, and contextual queries, plus end-to-end latency.

5.1 Setup

We sampled task episodes from AndroidControl spanning 10 application categories representative of common mobile use cases. To ensure balanced coverage, we sampled 5 episodes per category, yielding 50 episodes in total. For each episode, we applied a two-stage filtering pipeline. First, screens were retained only if they contained at least three distinct interactive options, excluded generic navigation controls (back, home, search), and were not login or confirmation dialogs. Second, from the qualifying screens within each episode, we selected three at fixed relative positions — the first, middle, and last — to represent the early, middle, and late stages of the task. This procedure yielded 150 screen states.

[1]Android Developers: https://developer.android.com/reference/android/view/accessibility/AccessibilityNodeInfo

Table 1. Summary of technical evaluation results. Detailed information is reported in Appendix A.1.

Agent	Evaluation Target	Accuracy	Latency (Mean)	Latency (SD)
Planning Agent	Subtask description	97.3%	5.89s	1.56s
Decision Agent	SCB identification	83.7%	13.68s	3.71s
Decision Agent	Top-most relevant element	88.3%	–	–
Decision Agent	Decision-support element	89.0%	–	–
Talk Agent	Contextual query answering	94.0%	5.24s	1.73s
End-to-end UI Update	Planning + Decision	–	~19.57s	-

Two researchers independently annotated each screen with: (1) the correct subtask description, (2) the classification of each UI element as SCB, top-most relevant, decision-support, and (3) answers to two standardized contextual queries. Each dimension was rated as correct, partially correct, or incorrect. Disagreements were resolved through discussion. Inter-rater reliability was high, with Cohen's $\kappa = 0.84$.

5.2 Results

Table 1 reports the overall accuracy and latency of FocalFlow's three-agent pipeline. We summarize the key findings below.

5.2.1 Accuracy. FocalFlow's three-agent pipeline achieved high overall accuracy, though the Decision Agent was more sensitive to visually complex or non-standard UI layouts. For example, lower SCB accuracy was observed in categories such as recipes, where content was often embedded in visual cards or dashboard-like layouts, while structured list-based interfaces (e.g., flights and weather) achieved better performance.

5.2.2 Latency. The pipeline re-runs whenever a user action causes the UI to update (e.g., double tapping or scrolling) and the latency reported in Table 1 corresponds to one such pipeline invocation. The *Decision Agent* accounts for approximately 70% of the total end-to-end UI update latency. Given that this stage jointly processes both the screenshot and the complete UI elements, which motivates future optimizations such as caching screen-level representations and reducing redundant UI-element analysis.

5.2.3 Error Analysis. Beyond generic LLM hallucination, we identified three recurring failure patterns that account for the majority of observed errors. Representative cases for each pattern are provided in Appendix A.1.2.

Modality mismatch. It arose when the accessibility tree and the visual screen disagreed. In some cases, the Planning Agent anchored subtasks to elements that remained in the tree but were visually occluded; in others, the Decision Agent overlooked symbolic controls that lacked content descriptions. This pattern is critical, as it causes FocalFlow's filtering mechanism to either fail completely or be misdirected. Such misinformation is exceptionally severe for BLV users, creating a direct barrier to task completion.

Semantic translation gaps. It appeared in icon-heavy interfaces, where developer-omitted accessibility metadata combined with the inherent ambiguity of abstract icons (e.g., a bordered-paragraph style button) caused both agents to settle on inconsistent textual interpretations. This pattern is also critical, as it may misdirect BLV users toward incorrect options. Consequently, users are forced to rely on the Talk Agent or iterative trial-and-error, both of which can be time-consuming.

Exploratory over-filtering. It emerged under open-ended goals: when the screen presented a set of equivalent candidate options that the user was expected to browse, the Decision Agent committed to a single top-most relevant element and demoted the rest to decision-support or irrelevant, narrowing the user's exploration space.

This pattern is recoverable: although not all options are surfaced, users can still complete the task, albeit with a potential cost to satisfaction.

Across the observed errors, modality mismatch and semantic translation gaps usually produced critical failures that fully blocked task progression, while exploratory over-filtering yielded recoverable inefficiencies that the user could resolve through Talk Agent queries or further exploration. Talk Agent errors were infrequent and largely consistent with generic hallucination. These patterns indicate that pipeline reliability is bounded by the quality of developer-provided accessibility metadata and the fragility of long-context multimodal inference.

6 USER STUDY

We conducted a within-subjects study with 12 BLV users to evaluate FocalFlow compared to a baseline. Specifically, we aimed to answer the following research questions:

(RQ1) **Effectiveness and Efficiency**: How does FocalFlow impact task performance and information access efficiency compared to the baseline?
(RQ2) **User Perception and Satisfaction**: How do BLV users perceive the usability, workload, and decision satisfaction when using FocalFlow?
(RQ3) **Decision-Making Patterns**: How does FocalFlow transform the decision-making heuristics and adaptive information-seeking behaviors of BLV users?

6.1 Participants and Apparatus

We recruited the same 12 BLV users who participated in the formative study. Each participant used two systems: FocalFlow and a baseline. The baseline was Google's TalkBack, augmented with interaction logging capabilities. Both systems were run on a Redmi Note 12T Pro. Each participant adjusted TalkBack's speech rate according to their preferences. Each participant received a compensation of approximately 28 USD. The study was approved by the Institutional Review Board (IRB) of our institution.

6.2 Procedure

We first collected participants' demographics and asked about their prior experiences with a screen reader. Before the main tasks, participants completed a 10-minute standardized familiarization session using a practice task (hailing a taxi to a massage salon) using both the baseline TalkBack and FocalFlow system. This session was designed to help them acclimate to the standardized experimental Android smartphone, the baseline TalkBack configuration, our logging setup, and the novel FocalFlow gestures.

Following the tutorial, the main comparison study commenced. Participants were required to use both systems to perform two multi-step tasks on mainstream applications that they did not regularly use: (1) ordering a hamburger for delivery via Meituan Waimai, and (2) scheduling a dermatology appointment at a hospital via Health 160. These tasks were selected to elicit the complex decision-making patterns identified in our formative study. The order of task and system was counter-balanced, and participants were asked to perform the tasks as they normally would in their daily lives and think aloud during the study. Each task was subject to a 15-minute hard time limit. Participants self-reported when they believed they had completed a task, and researchers then verified completion against a predefined task endpoint. A task failed on timeout (>15 min), early termination due to frustration, or self-report not matching the predefined endpoint. Upon completing the tasks with both systems, participants completed a questionnaire, with questions shown in Fig. 5. Finally, a semi-structured interview was conducted to gather more in-depth feedback.

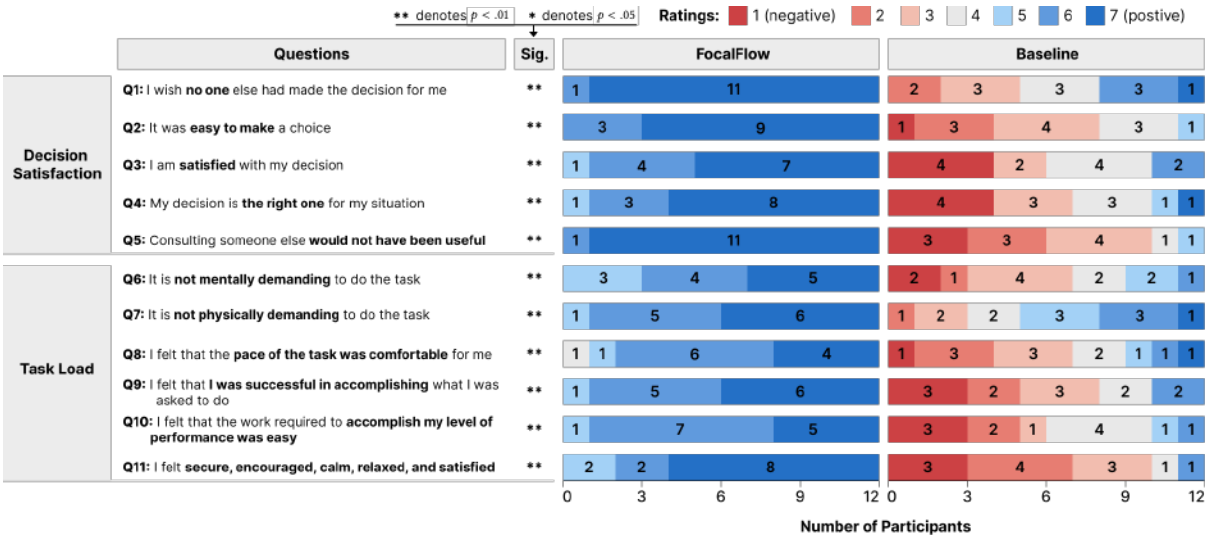


Fig. 5. Distributions of the ratings on Decision Satisfaction and Task Load for FocalFlow and the Baseline (1=strongly negative, 7=strongly positive). The asterisks indicate the statistical significance as a result of the Wilcoxon signed-rank test. Detailed statistics are provided in Table 3.

6.3 Metrics

To answer the aforementioned research questions, we collected a series of subjective and objective metrics to measure the effects of FocalFlow across four dimensions: system usability, task performance, decision quality, and user behavior.

- **System Usability.** We investigated the system's usability and user experience using selected questions from established questionnaires (i.e. SUS and NASA-TLX)[36, 59], all measured on 7-point Likert scales.
- **Task Performance.** We captured overall effectiveness and efficiency through Task Success Rate (number of completed tasks/total number of tasks) and *Task Completion Time* [72]. Task success was coded by the researchers according to whether the predefined task endpoint was reached. We also documented *the Frequency of Error Recoveries* required during task execution[34].
- **Decision Quality.** We employed the Decision Attitude Scale (DAS) to evaluate their satisfaction with the task outcomes [88], all measured on 7-point Likert scales. To quantitatively evaluate how effectively participants could discover relevant information, we also analyzed the *Exploration Coverage Rate*. This metric was calculated for each participant and task as follows:

$$R_{\text{coverage}} = \frac{\text{Number of SCB Explored}}{\text{Total Number of SCB}}$$

- **User Behavior.** We first calculated the *Distribution of Interaction Time* and *Distribution of Interaction Frequency* across three predefined on-screen element categories (Top-most Relevant, Decision-Support, and Irrelevant) to understand BLV users' attention allocation. We also explored the *Number of Requests for Decision-Support Information* to capture instances where the default, top-most relevant information was perceived as insufficient for making a confident decision.

6.4 Data Analysis

We collected data from multiple sources: audio and screen sharing recordings of the sessions, screen reader interaction logs, post-session questionnaire responses and interviews. All audio recordings and interviews were transcribed. To analyze the qualitative data, we employed a thematic analysis [17]. Specifically, two researchers first familiarized themselves with the data and independently performed open coding on the transcripts and interaction logs to identify patterns in participants' behaviors and feedback. We then compared the coding results and resolved discrepancies through iterative discussions until a consensus was reached. Finally, these codes were collated into potential themes and reported in the findings.

We assessed the normality of the difference scores for each quantitative measure using the Shapiro–Wilk test [91]. The *Exploration Coverage Rate*, *Task Completion Time*, *Interaction Time* and *Interaction Frequency* did not show significant deviations from normality and were compared between conditions using paired-samples t-tests [100]. Questionnaire responses, which were based on ordinal Likert-type scales, and the *Frequency of Error Recoveries*, which violated the normality assumption, were analyzed using the Wilcoxon signed-rank test [115].

To establish a reliable ground truth for *Exploration Coverage Rate*, two researchers independently analyzed the user interface for each task. They coded all SCBs in both conditions based on their relevance to the task's completion. Any discrepancies in coding were resolved through discussion until a consensus was reached, ensuring objectivity. A SCB was defined as having been explored if the screen reader's focus remained on it for a minimum of one second, as captured through our interaction logs. The interview audio was transcribed and thematically analyzed in alignment with our research questions [17]. Our findings are reported based on this analysis.

7 FINDINGS

In the following, we report the findings regarding the three RQs: effectiveness and efficiency, user perception and satisfaction, decision-making patterns.

7.1 Effectiveness and Efficiency (RQ1)

7.1.1 Task Completion Rate and Time Efficiency. The task completion rate was higher in the FocalFlow condition, where 11 of 12 participants (91.67%) were successful, compared to 8 of 12 (66.67%) in the baseline condition. Of the four participants who failed in the baseline condition, two exceeded the maximum allotted time, and two discontinued the task due to frustration. The single failure in the FocalFlow condition resulted from a modality-mismatch failure consistent with the pattern reported in Sec. 5.2.3.

For the time-on-task analysis, we excluded two participants who ended the task early due to frustration, one of whom also failed the task in the FocalFlow condition, resulting in a final sample of 10 participants. Mean completion time did not differ significantly between the baseline (M = 11.93min, SD = 3.03) and FocalFlow conditions (M = 12.29min, SD = 2.44; $t = -0.34$, $p = .74$), despite each on-demand voice inquiry in FocalFlow incurring an average end-to-end latency of 11.88s (STT + Talk Agent + TTS).

7.1.2 Distribution of Interaction Time and Frequency. As shown in Fig. 6A, our analysis of interaction time reveals the inefficiency of the baseline condition, where participants spent an overwhelming 90.85% of their time on *Irrelevant Elements*. Consequently, only a fraction of their effort was directed at task-related elements: 6.63% (SD=11.10%) on *Top-most Relevant Elements* and a mere 2.52% (SD=7.46%) on *Decision-Support Elements*.

In contrast, FocalFlow effectively directed user attention to task-related information. Participants spent a significantly greater proportion of their time on *Top-most Relevant Elements* (M=96.79%, SD=2.48%) compared to the baseline condition ($t = 31.47$, $p < .01$). Correspondingly, the time allocated to *Decision-Support Elements* was minimal (M=3.21%, SD=1.46%) and did not differ significantly from the baseline ($t = -1.51$, $p = .23$).

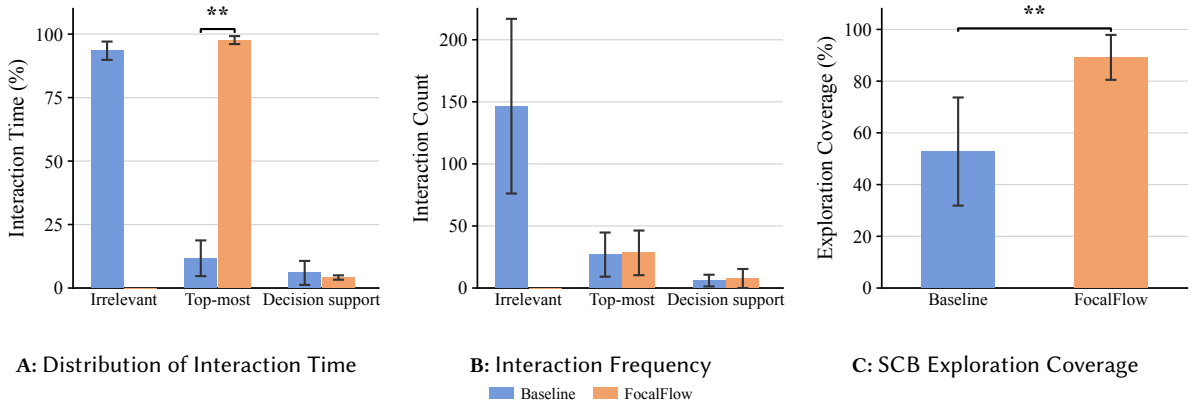


Fig. 6. Interaction behavior and exploration coverage across the Baseline and FocalFlow conditions. Panel A shows the distribution of interaction time across *Irrelevant Elements*, *Top-most Relevant Elements*, and *Decision-Support Elements*. Panel B shows interaction frequency across the same three categories. Panel C shows the SCB exploration coverage rate. In Panels A and B, the FocalFlow values for *Irrelevant Elements* are zero and therefore have no visible bars. Error bars indicate standard deviations. ** denotes $p < .01$.

As shown in Fig. 6B, participants in the baseline condition interacted overwhelmingly with irrelevant elements (M=146.50). This number far surpassed interactions with top-most relevant elements (M=26.92) and decision-support elements (M=6.00). Further comparison revealed that the interaction frequencies with relevant elements were comparable between the baseline and FocalFlow conditions. Specifically, we found no significant differences for either top-most relevant elements ($M_{Baseline} = 26.92$, $SD_{Baseline} = 17.79$ vs. $M_{FocalFlow} = 28.33$, $SD_{FocalFlow} = 17.99$; $p = .85$) or decision-support elements ($M_{Baseline} = 6.00$, $SD_{Baseline} = 4.71$ vs. $M_{FocalFlow} = 7.40$, $SD_{FocalFlow} = 7.95$; $p = 0.67$). These results suggest that while participants in both conditions engaged similarly with relevant information, the baseline experience was hampered by excessive and unproductive exploration of irrelevant content.

7.1.3 Exploration Coverage Rate. Our analysis revealed that the average exploration coverage rate for BLV users in the FocalFlow condition was significantly higher than that observed in the baseline condition ($t = -4.62$, $p < .01$), as illustrated in Fig. 6C. This finding indicates that FocalFlow enables BLV users to explore a wider range of available options, facilitating more informed decisions compared to the baseline. However, this finding also highlights a design challenge: participants tended to assume that any areas not presented by FocalFlow were non-existent. This observation aligns with the results presented in Sec. 7.3.3.

7.1.4 Frequency of Error Recoveries. **Participants under FocalFlow condition exhibited more intentional and exploratory navigation patterns.** Our analysis of user error recovery reveals that the frequency of error recoveries was significantly higher in the baseline condition than in the FocalFlow condition ($Z = -2.52$, $p < 0.05$). However, the underlying motivations for this behavior were different. In the baseline condition, participants used trial-and-error to recover from mistakes, such as backtracking after navigating to a wrong interface, which is consistent with previous research [16]. In the FocalFlow condition, error recovery was more exploratory. Participants proactively revisited previous subtasks not to correct errors, but to explore alternatives for a more satisfactory decision. For instance, P8, unsatisfied with the options in the "Hamburger" category, used the *Grounded Task Navigator* to return to the "Select Burger Category" subtask and chose to explore the "Juicy

Beef Burger" category instead. This suggests that FocalFlow may shift the role of error recovery from purely corrective actions toward a more flexible exploration of task paths, potentially supporting more informed and satisfied decision-making (further reported in Section 7.2.2).

7.2 User Perception and Satisfaction (RQ2)

7.2.1 User Perception. **FocalFlow was particularly praised for its ability to allocate users' cognitive resources to task-relevant activities.** Participants reported that using FocalFlow was a more intuitive experience. One participant found it "*a lot easier*" (P5), while another (P8) expressed enthusiasm, stating, "*it was such a surprise when I first felt that I just simply swiped (left or right) and it (FocalFlow) moved my focus to where I wanted it to be.*" This view was shared by numerous participants (P2, P4, P5, P7, P11), who specifically commended FocalFlow's ability to "*skip right over elements that had nothing to do with what I wanted to do,*" P12 recalled, "*It is not the kind of ease that requires no thought at all. On the contrary, I do feel tired, but that is because I am deeply engaged in thinking about which option I truly want.*" Participants contrasted FocalFlow's deliberate decision-making support with the baseline, which P8 likened to "sorting garbage" due to the overwhelming presence of irrelevant information.

7.2.2 Decision Satisfaction. **FocalFlow effectively helps different BLV participants weigh their options according to their individual preferences.** Participants expressed significant satisfaction with their final choice ($Z = 3.06$, $p < .01$) and were significantly confident that it was the right decision for their situation ($Z = 2.93$, $p < .01$). For instance, while most participants chose KFC for their hamburger orders, P3, after weighing the delivery fees, opted for Tastien, which offered free delivery. In contrast, under baseline conditions where the cost of information gathering was higher, participants tended to simply choose the first option they discovered. They justified this by saying it was "the one I happened to hear of" (P3, P8, P10) and that "looking for others would be too much trouble." (P1, P8, P11) Furthermore, the decision-making process was perceived as significantly easier ($Z = 3.06$, $p < .01$). Participants also reported a strong sense of personal agency over their decision ($Z = 2.93$, $p < .01$) and felt FocalFlow provided sufficient information to make external consultation unnecessary ($Z = 3.06$, $p < .01$).

7.2.3 Perception of System Latency. **All participants agreed that AI response latency impacted their willingness to use FocalFlow in the future.** Participants thought the pace of the task was significantly more comfortable (Z=3.22, p<.01), but unanimously identified the API request time as a major drawback, describing the wait as "*so boring*" (P1, P3), "*almost putting me to sleep*" (P9, P10, P11) and "*I'd rather do it myself*" (P4), this finding is consistent with previous research[53]. Despite this, participants expressed a higher willingness to use FocalFlow frequently ($M = 4.75$, $SD = 2.00$) compared to the baseline ($M = 3.75$, $SD = 1.48$). They suggested FocalFlow's utility was situational, being particularly helpful for "*complex and unfamiliar tasks*" (P6, P8, P9) but "*not necessary for some easy tasks such as instant messaging*" (P5).

7.3 Decision-Making Patterns (RQ3)

This section details the decision-making patterns observed with FocalFlow, specifically examining the shift in decision heuristics (7.3.1), the adaptation of information gathering strategies based on familiarity (7.3.2), the mechanisms for trust calibration regarding information completeness (7.3.3), and the evolving preferences for system adaptability (7.3.4).

7.3.1 Decision-Making Heuristics. **FocalFlow facilitates a transition in the decision-making behaviors from a "*satisficing*" heuristic to an "*optimizing*" strategy centered on personal utility.** Regarding the Baseline system, participants frequently employed terms such as "tiresome,"(N=7) "exhausting,"(N=5) or "painful"(N=3) to describe the interaction process. Consequently, many reported adopting a "*get it over with*"

(N=3) strategy, explaining that they would "*settle for the earlier okay options*" (N=10) rather than striving for the optimal one. This aligns with the lower satisfaction scores and shorter dwell times observed previously.

With FocalFlow, however, the participants' feedback shifted toward deliberation and preference. Participants reported that because the exploration felt "*effortless*" (N=7), they were willing to "*engage in more complex evaluations of trade-offs*" (P7, P12). For example, P12 prioritized transportation accessibility over physical distance, explicitly stating a preference to avoid *"walking outside for too long"* despite the existence of closer options. This suggests that under conditions of reduced cognitive load, BLV users are not just seeking to complete a task, but are proactively optimizing for their personal utility.

7.3.2 Information Gathering and Verification Processes. **FocalFlow shifted BLV participants toward a "*check-and-move*" (N=1) or "*layer-by-layer*" (N=4) strategy, moving away from the "*lottery-like*" (N=3) passive exploration typical of traditional screen readers.** Under the baseline, lacking semantic filtering, participants repeatedly committed to UI elements that sounded topically related but were in fact unrelated to the current task. For example, under the baseline, the information P3 encountered first shaped what he thought the page was for. As P3 recalled, "At first, I heard a lot about symptoms, so I thought the page was for narrowing down a condition. It was only after I swiped further and heard a hospital name that I realized I could also choose a hospital on the same page." Participants could spend time on information unrelated to their current task without knowing whether they had reached everything needed. Under FocalFlow, however, participants rarely committed to the first option encountered. Instead, they dwelled on a subset of two-to-three candidates, repeatedly triggered On-demand Contextual Expansion to surface decision-support attributes (e.g., distance, transportation), and re-listened to the same option's expanded details before making a decision. As P12 put it: "I kept listening to the rating and the doctor's specialty for each one. I wanted to be sure before I picked." The same contrast appears in the earlier results: participants spent more time on irrelevant elements with TalkBack (Sec. 7.1.2), but explored more of the available task information with FocalFlow (Sec. 7.1.3).

FocalFlow helped participants anchor their progress by situating their current state within the task flow, effectively mitigating the uncertainty common in multi-step transitions. Rather than relying on tedious manual exploration to orient themselves, participants used the system-generated summaries to confirm that the interface matched their expectations. As P5 highlighted, *"it (Grounded Task Navigator) not only tells me where I am, but also lets me know if I did the last step right."* This immediate validation allowed participants to proceed to subsequent decisions with the confidence that their "previous action had been correctly executed" (P3, P12). This suggests that FocalFlow facilitates the management of complex task flows by providing an explicit confirmation of progress at each step.

7.3.3 Trust Calibration and Perception of Information Completeness. **Participants expressed concerns regarding the comprehensiveness of the filtered information and actively employed verification strategies to mitigate the risk of missing critical options.** Although the dynamic filtering reduced clutter, several participants (N=3) worried that the AI might filter "*too aggressively*" (P3), potentially "*hiding valid options*" (P2, P3, P9). This skepticism manifested as a desire to confirm the absence rather than just the presence of elements. For instance, while some participants (N=3) eventually felt "*reassured*" (P12) after the AI explicitly confirmed that there was no other information, others maintained a lingering "*disbelief*" (P1, P7). These participants described a discrepancy between their expectations and the auditory feedback, P7 remarking, "*I feel like there should be something here, but I didn't hear it (FocalFlow) mention this category,*" even when such elements were genuinely absent from the screen. However, all participants (N=12) agreed that having *On-demand Contextual Expansion* as a "*source of reassurance*" (P8) alleviated the cognitive burden of not knowing the full interface state. These findings highlight that for BLV users, the ability to definitively verify the non-existence of expected features is sometimes more valuable than merely identifying existing content.

This skepticism was further validated by instances where the AI struggled with implicit domain knowledge. P1 noted a specific failure case where her preferred choice was excluded: "*I know there's a pizza chain that makes killer burgers, but the AI left it off the list. It probably just figured a pizza store wouldn't have them since I specifically asked for 'burgers'*" This suggests that AI's inability to capture implicit connections may inadvertently narrow the user's decision space.

7.3.4 User Preferences for Adaptability. **Participants desire an adaptive system that aligns information presentation and automation levels with their personal habits and situational contexts.** Some participants (N=3) explicitly expressed a need for the system to adapt to their established information-seeking habits. For example, price-sensitive participants, like P3, suggested, "*I wish it would just tell me that straight away without me asking.*" While others (N=9) prefer to *keep it simple* (P8) , and noted: "don't give it to me unless I need it" (N=5). Furthermore, the demand for automation fluctuated based on physical context. While manual confirmation was appreciated in quiet settings, some participants (N=3) expressed a need for higher automation when mobile. As P5 explained, "*When I am walking outdoors with my cane, I don't want to swipe. I just want it to select for me so I don't have to stop.*"

8 DISCUSSION

8.1 Supporting Deliberate Decision-Making in Non-Visual Contexts

8.1.1 From Linear Scanning to Task Progress Anchoring. Our findings highlight that the frequent full-screen transitions typical of mobile scenarios impose a unique challenge compared to the persistent layouts of web interfaces. For BLV users, these transitions often result in fragmented interactions and high uncertainty regarding task continuity [2, 73]. While traditional navigation requires a laborious bottom-up approach-relying on linear scanning to reconstruct the interface state after each transition [14]-our results suggest an opportunity to transcend simple element-level announcements. FocalFlow addresses this by providing functional summaries that situate the current state within the global task flow; the system serves as a grounding mechanism that validates past actions and facilitates immediate problem identification [24]. This approach not only mitigates the cognitive burden of re-orientation but also explicitly confirms the user's status, enabling them to anticipate forthcoming steps and proceed with higher confidence.

8.1.2 Reallocating Cognitive Resources: From Filtering to Evaluating. To understand the trade-offs observed in our study, we first examine the baseline behavior through the lens of information foraging theory [77, 105] and bounded rationality [94]. In the traditional screen reader condition, participants exhibited significantly higher interaction frequencies and longer interaction times on irrelevant elements. This aligns with the classical foraging model, where BLV users expend cognitive effort following information scents and filtering out noise to locate valuable patches of information. Consequently, the baseline navigation left fewer resources available for the actual decision-making task. Crucially, this high interaction cost often forces users into a "*satisficing*" heuristic [95]. As reflected in our findings, the process was perceived as "*tiresome*" or "*exhausting*," leading users to adopt a strategy of settling for early "*good enough*" options to terminate the interaction [92].

FocalFlow, however, reconfigures this workflow by intervening at two distinct levels. First, it reduces the physical cost of foraging by providing relevant information automatically. Instead of manually navigating through low-scent patches, users delegate this navigational labor to the AI, which acts as a proxy forager. Second, at a cognitive level, FocalFlow facilitates a form of distributed cognition [38], where the burden of filtering is offloaded to the AI agent. Our findings indicate that users reinvested these freed resources to shift from merely seeking task completion to proactively optimizing for personal utility. Participants engaged in complex evaluations of trade-offs, such as weighing transportation accessibility against physical distance, demonstrating behaviors characteristic of an "*optimizing*" strategy rather than a sub-optimal compromise. This delegation allows

users to pivot their cognitive resources from low-level information gathering to high-level content evaluation. Consequently, FocalFlow enables BLV users to make more deliberate decisions while reducing their cognitive load.

8.2 Extending to Real-World Scenarios

8.2.1 Deployment. FocalFlow's laptop-hosted Python backend runs the multi-agent pipeline, while a cloud LLM API provides inference. Reimplementing the pipeline as an Android service could remove the laptop and ADB connection while retaining the same cloud API. However, the privacy concern would remain because interface data would still be sent to the LLM provider.

Fully on-device deployment would instead run the agent backend and perform model inference on the phone, potentially drawing on smaller, mobile-targeted models such as Gemma 4 E2B/E4B [32] and Phi-3-mini [1]. This would keep interaction data and task history on the device. However, running the full pipeline within a phone's memory and energy budgets may require smaller models or shorter context windows, potentially weakening cross-screen reasoning and silently narrowing the options surfaced to BLV users. These trade-offs motivate a hybrid design: routine subtask classification and single-screen filtering could remain local, while cloud inference could handle cross-screen comparisons that remain uncertain or require broader domain knowledge. Such a design should preserve task state when work moves between local and cloud execution, make cloud use visible to BLV users, and let them approve or decline data transfer. Future studies could compare whether each configuration still surfaces the options users need and whether users retain control over when data leave the phone.

8.2.2 Privacy. On each invocation, FocalFlow transmits a pruned accessibility tree (text and contentDescription with element identifiers) plus the current screenshot to the LLM provider. This is narrower than the full view hierarchy but still carries substantial personal data. Informed consent is structurally harder here than in mainstream AI assistance. A sighted user of a vision-based assistant can visually preview what they send, but a BLV user cannot. Once FocalFlow is enabled, screen contents flow continuously to the model with no low-cost way for the user to verify each transmission. Prior work has surfaced related concerns in BLV use of generative AI [4], but the issue compounds when AI mediation is the default rather than an explicit, single-shot invocation. Brief auditory cues at transmission might partially mitigate this. Any architecture delegating UI interpretation to a remote model inherits its privacy cost. We treat this as a structural property of the design space and one that warrants sustained attention as LLM-based assistance enters routine accessibility deployment.

8.2.3 Authority and Accountability. A subtler concern is what authority transfers when an AI mediates between an app and a BLV user. With native screen readers, app developers have traditionally controlled what is presented and how, but FocalFlow's *Decision Agent* can demote, reorder, or suppress these authored choices. A fully WCAG-compliant screen still passes through a filter before reaching the user. At the same time, BLV users have traditionally held the relevance judgment themselves through sequential scanning, and FocalFlow performs that judgment first, presenting only its result. This transfer raises two limitations. First, filter errors are silent: when the *Decision Agent* demotes a relevant SCB, the omission produces no audible anomaly, unlike a misread label or an unlabeled icon. The *Talk Agent* and error recovery exist, but invoking them presupposes the user already suspects something is missing. Second, verification cost is regressive: a sighted user of a vision-based assistant can visually spot-check the AI at near-zero cost, while a BLV user verifying FocalFlow's filter must traverse the unfiltered tree manually — the very labor FocalFlow was designed to reduce. Questioning the system thus requires temporarily forgoing its benefit. Silent errors call for evaluation methods that go beyond task completion to surface what users did not realize they missed. Regressive verification cost calls for interaction patterns where questioning the system does not unwind its benefit, an open design problem with no clean mainstream analogue.

Both directions are concerns that must be addressed alongside the accessibility community's efforts to bring AI's benefits to BLV users.

8.3 Design Implications

While FocalFlow demonstrates the potential of LLM-powered agents to support BLV users' decision-making, our evaluation uncovered inherent tensions in AI-mediated accessibility. Beyond the specific implementation of FocalFlow, these findings point to broader design implications for creating robust non-visual interaction agents in the future.

8.3.1 Bridging the Semantic Gap. Our study reveals that for BLV users, decision-making relies not only on the literal UI hierarchy but also on implicit semantic connections, such as the burger-pizza mismatch (Section 7.3.3) and the single task failure in Section 7.1.1 stemmed from an LLM hallucination where the model failed to recognize a SCB. These cases suggest that relying solely on the UI view hierarchy might not be sufficient for complex reasoning. The AI acted as a literal interpreter of the screen rather than a knowledgeable assistant. To address this gap, future systems could consider transcending simple UI scraping, such as a hybrid knowledge architecture that augments the LLM's visual understanding with external knowledge bases (e.g., retrieving restaurant menus or domain-specific knowledge) to capture implicit connections[11, 108]. Moreover, incorporating multimodal capabilities enables the agent to perceive UI elements through their visual and functional context rather than relying solely on textual metadata. For instance, by inferring that an element nested within a category list is likely a related option, the model might mitigate hallucinations caused by misidentified controls.

8.3.2 Balancing Efficient Exploration with Trust Calibration. Curating information to reduce cognitive load inherently positions the AI as an intermediary that controls information access [4]. Our findings confirm that FocalFlow empowered BLV users to explore a significantly wider range of task-relevant options compared to the baseline. While minimalist delivery of information can enhance clarity and reduce mental effort [69], it also introduces a nuanced challenge regarding trust calibration. As observed in Section 7.3.3, the high reliability of the curation led some participants to perceive the presented outcome as exhaustive and cease to verify whether important information was omitted. This indicates that the black-box nature of filtering can unintentionally discourage users from questioning the system's boundaries. To mitigate this, future designs should prioritize transparency in the curation process. This could be achieved through subtle auditory cues for filtered content or a "*full exploration*" mode that allows users to bypass filters and access all interface elements. These could preserve the efficiency benefits of noise reduction while signaling to the user that more information is available if needed, thereby supporting a more informed and verified decision-making process.

8.3.3 Balancing Latency and Control through Adaptive Interaction. Our findings reveal that user preference for automation is governed by a trade-off between perceived utility and interaction cost. While participants expressed frustration with API latency for simple tasks (e.g., opening an app), they tolerated these delays for complex and unfamiliar tasks where the AI's utility outweighed the wait time. This also aligns with empirical evidence from assistive navigation systems [5] and from AAC contexts, where AI-mediated interfaces similarly trade user authorship for timeliness in social interaction[113]. This suggests that a one-size-fits-all approach to AI-assisted information curation is insufficient for diverse real-world contexts.

Future research on non-visual decision-making should consider context-aware hybrid architectures. Rather than routing every interaction through a uniform model, future systems could dynamically optimize the latency-capability trade-off by offloading time-sensitive tasks to lightweight on-device models, while reserving cloud-based inference for complex reasoning, or using multiple models to optimize latencies [20]. Also, such adaptability could extend to the physical volatility of mobile scenarios. Future designs could incorporate environmental sensing to maintain output intelligibility, such as dynamically adjusting speech parameters in response to ambient

noise [6]. Furthermore, adaptability frameworks should extend beyond physical context to account for individual information-seeking strategies. Since preferences for proactivity might vary among participants (e.g., price-sensitive users), future designs could leverage user modeling to dynamically tailor the presentation of top-most relevant information.

8.3.4 Scope and Boundary Conditions of Task-Oriented Filtering. The applicability of FocalFlow's filtering and guidance is bounded by a property of the task, not of the interface: it presupposes that user intent is stable enough for the system to anchor to. This holds for the multi-step decision tasks we evaluated, but the assumption weakens in activities where exploration itself is part of the cognitive process, such as open-ended browsing or data analysis.

Our three features differ in their sensitivity to this condition. The Grounded Task Navigator operates at the granularity of "where am I and what can I do next on this screen," and this grounding retains value whether a user is pursuing a fixed objective or moving through an interface with looser intent: a screen overview always orients, and a single-screen next-step suggestion can serve as a sensible default that the user is free to override. On-demand Contextual Expansion is likewise broadly applicable, since the user retains authorship over what to inquire about. Dynamic Focus Scoping, in contrast, is the feature most tightly coupled to a defined task trajectory. When user intent is genuinely fluid or constructed through interaction, filtering elements against a putative subtask risks foreclosing the very exploration that constitutes the task. Yet this coupling is a design choice — a relaxed variant that suppresses only categorically noisy content (e.g., advertisements, promotional banners) could support free exploration without imposing goal-directed structure.

We therefore position the full FocalFlow pipeline as best suited to goal-directed multi-step tasks, while noting that its orientation and on-demand mechanisms — together with a less restrictive form of focus scoping — could extend the approach's value to less structured activities.

9 LIMITATIONS AND FUTURE WORK

This work has several limitations that point to opportunities for future research.

First, human intent is often constructed dynamically [101], users need the flexibility to pivot their overarching goals. To fully support the fluid nature of human decision-making, future systems should evolve from rigid plan execution to negotiated agency. This implies designing AI agents that act as a collaborative partner that supports fluid goal renegotiation and makes the boundaries of its curation explicit to the user.

Second, FocalFlow's generalizability is bounded by the semantic quality of the accessibility tree. FocalFlow identifies an element's function from its *text* and *contentDescription* attributes. When developers omit these, exposing elements as *Unnamed Elements*, the system cannot reliably classify SCBs or provide meaningful action points. This constraint reflects a broader developer-side accessibility compliance issue affecting all assistive systems built on accessibility APIs, and motivates future work on vision-based fallbacks that infer element semantics directly from the screenshot when tree-level metadata is missing.

Third, FocalFlow's per-event re-invocation strategy makes the system poorly suited to interactions involving continuous scrolling, where each scroll event would trigger a full pipeline pass. Future work should explore incremental accessibility-tree updates that re-process only changed regions, and caching screen-level representations across small viewport shifts.

10 CONCLUSION

Modern mobile apps present accessibility barriers for BLV users. Screen readers linearize visual interfaces, causing cognitive overload from irrelevant information and complicating decision-making in multi-step tasks. Existing tools fall short, lacking the contextual awareness required for such complex scenarios. To bridge this gap, we introduced FocalFlow, an LLM-powered system that enhances screen readers with three features: a *Grounded*

Task Navigator for task orientation and error recovery; *Dynamic Focus Scoping* to reduce clutter while preserving the interface layout; and *On-demand Contextual Expansion* to provide critical details without overwhelming the user. Our user study evaluated that FocalFlow enables more deliberate, satisfactory decisions with less cognitive load. This work demonstrates the potential of LLM-driven information curation to enhance the accessibility of complex mobile workflows.

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A APPENDIX

A.1 Technical Evaluation Details

A.1.1 Detailed Per-Category Correctness Rates and Latency Statistics. Table 2 reports the per-category correctness rates across the three-agent pipeline.

Table 2. Per-category accuracy across the three-agent pipeline. The evaluation set comprises 10 categories, with 15 screens per category (5 episodes × 3 screen states per episode), totaling $N = 150$.

Category	Subtask	SCB	Top-Relevant	Decision-Support	Contextual Query
Email	96.7%	90.0%	93.3%	90.0%	93.3%
Flights	100.0%	100.0%	96.7%	100.0%	100.0%
Health & Fitness	93.3%	70.0%	76.7%	80.0%	90.0%
Maps	100.0%	86.7%	93.3%	96.7%	96.7%
Finance	100.0%	93.3%	96.7%	100.0%	96.7%
Notes & Todos	100.0%	90.0%	96.7%	96.7%	96.7%
Productivity	93.3%	76.7%	83.3%	86.7%	93.3%
Recipes	100.0%	60.0%	70.0%	66.7%	86.7%
Shopping	90.0%	70.0%	76.7%	73.3%	86.7%
Weather	100.0%	100.0%	100.0%	100.0%	100.0%
Overall	**97.3%**	**83.7%**	**88.3%**	**89.0%**	**94.0%**

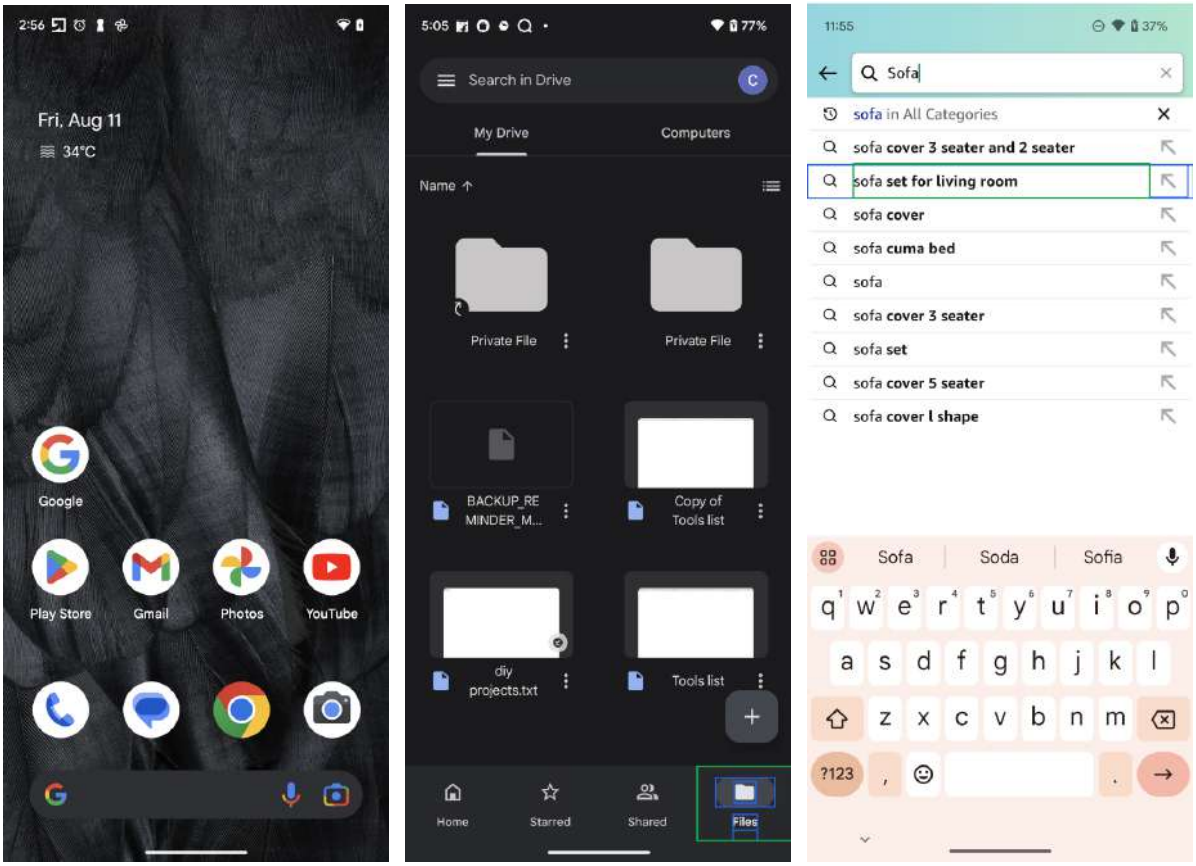
Note: The "Contextual Query" column evaluates the Talk Agent's performance on two standardized contextual queries:
(1) "What categories of additional information are available on this screen?"
(2) "Can you describe the differences between the first and second option?"

A.1.2 Representative Failure Cases. We illustrate each of the three failure patterns described in Section 5.2.3 with a representative case from the AndroidControl evaluation. In each annotated screenshot, green boxes denote top-most relevant elements, and blue boxes denote decision-support elements.

Case A — Modality mismatch (Fig. 7a). The Planning Agent issued the subtask "Open the Yahoo Mail app" on the Android Launcher home screen. While the visible viewport exposed only ten app icons, the accessibility tree returned by the system additionally enumerated nodes belonging to the adjacent home-screen page — including the Yahoo Mail icon, which was not rendered in the current viewport. Treating the tree as ground truth, the Decision Agent elevated this off-screen Yahoo Mail node as the top-most relevant element. The VAT thus directed the screen reader's focus traversal to a node that had no on-screen counterpart. We classify this as a critical failure, since recovery requires the user to independently infer that a horizontal swipe of the home screen is needed to bring the announced target into the active modality.

Case B — Semantic translation gap (Fig. 7b). The Planning Agent issued the subtask "Enter the Trash to view deleted files" on the Google Drive My Drive view. The correct entry point is the menu button at the top-left, which opens a navigation drawer containing the Trash item. Instead, the Decision Agent elevated the bottom-bar Files tab as the sole top-most relevant element, conflating two controls that share surface-level lexical proximity but play unrelated navigational roles. The menu icon and the Trash item it would reveal were demoted to Irrelevant. The user is silently routed toward a tab that does not satisfy the subtask and recovery requires either a Talk Agent query or manual exploration, inflating time-on-task, so we classify this as a critical failure.

Case C — Exploratory over-filtering (Fig. 7c). The Planning Agent issued the subtask "Select a specific search suggestion" on the search-suggestions overlay (query: "sofa"). The screen presented nine roughly equivalent



(a) Case A: Modality mismatch. (b) Case B: Semantic translation gap. (c) Case C: Exploratory over-filtering.

Fig. 7. Representative failure patterns in the AndroidControl evaluation. Annotated screenshots show green boxes (top-most relevant elements), and blue boxes (decision-support elements).

autocomplete suggestions, each a valid next step for an open-ended browsing intent. The Decision Agent committed to a single suggestion ("sofa set for living room") as top-most relevant and demoted the remaining eight to irrelevant, collapsing a parallel decision space into a single-option task. The user is steered toward one suggestion without being made aware that eight comparable alternatives exist on the same screen. Recovery requires the user to actively suspect that filtering has occurred. We classify this as a recoverable error.

A.2 Other Details

Table 3. Detailed statistics of subjective ratings from the user study (1 = strongly negative, 7 = strongly positive). SD denotes the standard deviation. Significance was analyzed using the Wilcoxon signed-rank test.

Aspects	Questions	FocalFlow		Baseline		Significance	
		Mean	SD	Mean	SD	Z	p
Decision Satisfaction	Q1: I wish no one else had made the decision for me.	6.92	0.29	3.92	1.44	2.93	$< .01$
	Q2: It was easy to make a choice.	6.75	0.45	3.00	1.13	3.06	$< .01$
	Q3: I am satisfied with my decision.	6.50	0.67	3.17	1.85	3.06	$< .01$
	Q4: My decision is the right one for my situation.	6.58	0.67	3.08	1.88	2.93	$< .01$
	Q5: Consulting someone else would not have been useful.	6.92	0.29	2.58	1.44	3.06	$< .01$
Task Load	Q6: It is not mentally demanding to do the task.	6.17	0.83	3.33	1.56	3.09	$< .01$
	Q7: It is not physically demanding to do the task.	6.42	0.67	4.67	1.50	3.01	$< .01$
	Q8: I felt that the pace of the task was comfortable for me.	6.08	0.90	3.50	1.78	3.22	$< .01$
	Q9: I felt that I was successful in accomplishing what I was asked to do.	6.42	0.67	3.00	1.76	2.93	$< .01$
	Q10: I felt that the work required to accomplish my level of performance was easy.	6.25	0.62	3.08	1.68	3.06	$< .01$
	Q11: I felt secure, encouraged, calm, relaxed, and satisfied.	6.50	0.80	2.50	1.45	3.08	$< .01$
System Usability	Q12: I think that I would like to use this system frequently.	4.75	2.01	3.75	1.48	1.08	$= 0.28$
	Q13: I did not find the system unnecessarily complex.	6.17	1.03	3.92	1.73	2.80	$< .01$
	Q14: I thought the system was easy to use.	6.33	0.65	4.17	1.03	2.81	$< .01$
	Q15: I think that I do not need the support of a technical person or sighted people to be able to use this system.	6.58	0.51	4.17	1.59	2.46	$< .01$
	Q16: I found the various functions in this system were well integrated.	5.58	1.08	4.58	1.44	2.67	$< .01$
	Q17: I thought there was not much inconsistency in this system	6.33	0.89	4.67	1.78	2.93	$< .01$
	Q18: I would imagine that most people would learn to use this system very quickly.	6.17	0.72	3.58	1.24	2.47	$< .01$
	Q19: I did not find the system very cumbersome to use.	6.33	0.98	4.67	1.37	2.93	$< .01$
	Q20: I felt very confident using the system.	6.58	0.51	5.42	1.31	2.12	$< .05$
	Q21: I do not need to learn a lot of things before I could get going with this system.	6.00	1.21	3.50	2.02	2.39	$< .01$

Table 4. BLV participants' demographic information. All vision conditions were self-reported during recruitment screening.

ID	Gender	Age	Vision level	Functional vision	Age of onset	Experience (years)	Device experience (years)
1	M	27	Totally blind	Light Perception	5	10	3 (Android), 7 (iOS)
2	F	25	Totally blind	Light Perception	1	10	10 (Android)
3	F	34	Totally blind	Light Perception	3	10	3 (Android), 7 (iOS)
4	M	28	Low vision	Counting Fingers	5	3	3 (iOS)
5	M	36	Totally blind	Light Perception	1	12	10 (Android), 2 (iOS)
6	M	43	Totally blind	Light Perception	3	11	10 (Android), 1 (iOS)
7	F	24	Low vision	Form Perception	1	10	10 (Android)
8	F	28	Totally blind	Light Perception	1	11	8 (Android), 3 (iOS)
9	F	23	Low vision	Form Perception	9	10	3 (Android), 7 (iOS)
10	F	28	Totally blind	Light Perception	1	11	11 (Android)
11	F	33	Totally blind	Light Perception	2	10	3 (Android), 7 (iOS)
12	M	34	Totally blind	No Light Perception	9	11	3 (Android), 8 (iOS)

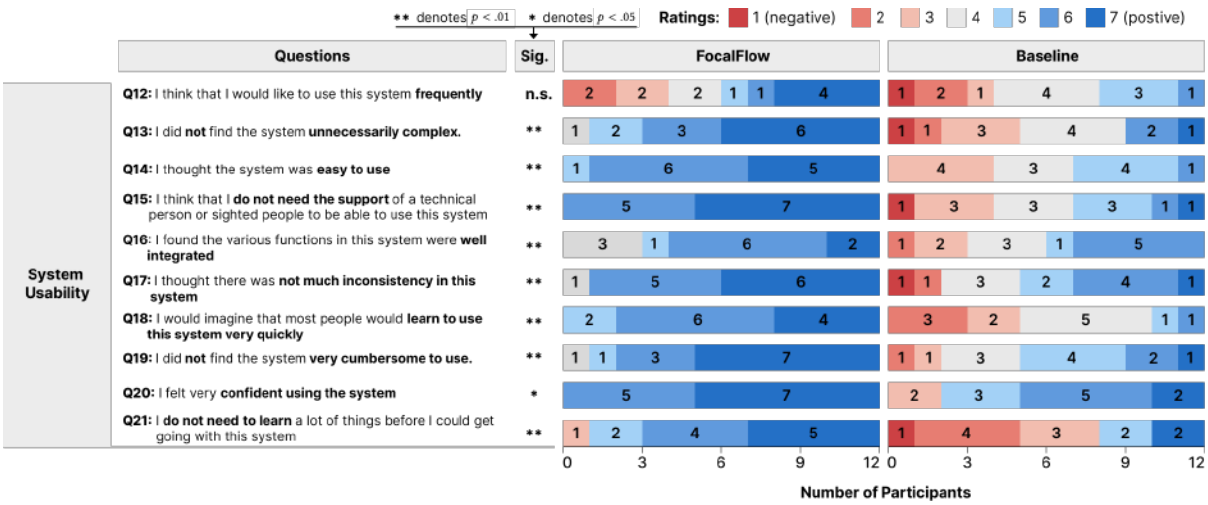


Fig. 8. Distributions of the ratings on System Usability for FocalFlow and the Baseline (1=strongly negative, 7=strongly positive). The asterisks indicate the statistical significance as a result of the Wilcoxon signed-rank test. Detailed statistics are provided in Table 3.

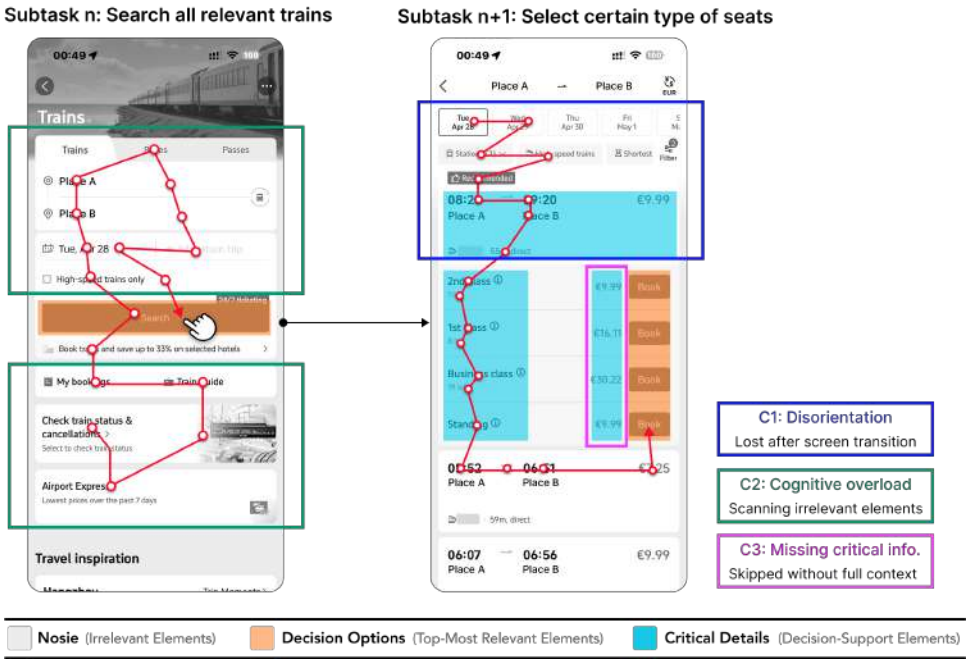


Fig. 9. The complete task flow of user using a traditional screen reader to book a train ticket. The red navigation path illustrates the inefficiency of linear scanning. The visualization highlights exactly where participants encountered the three primary barriers identified in our study: disorientation following screen transitions (C1), cognitive overload from sifting through irrelevant UI elements (C2), and the risk of committing to actions without hearing fragmented, critical details (C3).

A.1 Prompts for Planning Agent

1. Task Background and Context User's Ultimate Goal: instruction
Historical Context: To understand the current state, review the sequence of previously completed subtasks. The next subtask must be a logical continuation of this history.
...
2. Input Information You are provided with the following information to analyze the current screen: UI Element Data (Ground Truth) This is the definitive, structured data of all interactable UI elements on the current screen. Treat this information as the single source of truth. Each element has four key attributes: name: The resource ID or unique identifier of the node. text: The text content displayed on the UI element. contentDescription: The accessibility text description of the UI element. children: The direct child nodes of the element, reflecting the UI's hierarchical relationship.
3. Your Core Task and Guidelines Your core task is to analyze all the provided information and define the next subtask required to achieve the user's ultimate goal. When defining this subtask, you MUST strictly adhere to the following guidelines:
Focus on Intent, Not Operation (Most Important): The subtask description MUST be a general user intent. It must describe what the user "wants to do" at a high level, not the specific UI operation they will perform.
Correct Examples: Select a store, Enter the checkout page, Search for a specific product.
Incorrect Examples: Click the "KFC" button, Click the shopping cart icon, Type "hamburger" in the search box.
Single Intent Principle: Each subtask should contain only one core user intent. DO NOT combine multiple distinct intents into a single subtask.
Correct Example: Select a car model. (The next task might be Call a vehicle.)
Incorrect Example: Select a car model and then call a vehicle.
Follow History, Step-by-Step: The new subtask MUST be a logical continuation of the historical steps. Skipping necessary steps is not allowed. If a required precursor action (e.g., Select a city) is not in the history, your next subtask MUST be Select a city, even if the screen appears to have a city pre-selected. If you have already completed a step, do not repeat it (e.g., if you have already chosen a hospital, do not choose a hospital again).
Handle Multiple Valid Paths (OR Condition): If the current screen offers several different but equally logical paths to advance the user's goal, you MUST identify these parallel possibilities. Frame the subtask as a choice using 'OR'.
Correct Example: Select a food category OR select a specific restaurant.
Incorrect Example: Select a food category and a restaurant.
4. Output Format Your output MUST strictly include the following three sections in English. DO NOT add any extra comments or explanations outside of these sections. Thought Please provide a detailed analysis of your decision-making process in English. Combine the user's goal, screen information, and historical context to deduce the next logical subtask. If there are multiple valid paths, explain why each is a reasonable choice and how they relate to the overall goal. Explain how your deduction conforms to the guidelines.
Subtask Based on your analysis, provide a brief, intent-focused description of the next subtask in English. If multiple valid paths exist, present them as a choice using 'OR'. Screen Provide a single, concise sentence in English describing the main purpose of the current screen. DO NOT mention specific UI elements, layouts, or operations.

A.2 Prompts for Decision Agent

1 ˙Task Background and Context
- **User's Ultimate Goal**: 'instruction' - **Historical Memory**: In previous operations, you recorded the following important information: 'memory'
2İnput Information
You are provided with two primary sources of information to understand the current screen:
1 ˙UI Element Data (Ground Truth)
This is the definitive, structured data of all interactable UI elements on the screen. Treat this information as the **single source of truth** for what elements exist and what text they contain. Each element contains four properties:
- 'name': The resource ID or unique identifier of the node.
- 'text': The text displayed on the element.
- 'contentDescription': The accessibility text description of the element.
- 'children': A list of direct child elements, reflecting the UI's hierarchical relationship.
<!– end list –>
2˙Screenshot (Visual Context)
A screenshot of the current page is provided alongside this data. Its purpose is to provide **visual context** for how the elements in the JSON are arranged and styled. Use the image to understand:
- **Spatial Layout and Grouping**: Crucially, use the screenshot to identify visually distinct groups, such as cards, list items, or dialog boxes.
3 ˙Current Subtask
Your immediate objective is: **current_subtask**
3 ˙Your Task and Analysis Criteria
Your core task is: **Using the Visual Context from the screenshot, identify the self-contained 'card' or 'list item' that is most relevant to the 'Current Subtask'. Then, output the unique 'name' of the minimal container element that acts as the root for that entire card.**
When analyzing, you must strictly adhere to the following criteria:
1. **Visually Identify the Card First**: Your primary analysis should start with the screenshot. Look for visual containers (cards, rows, panels) that hold information related to the subtask.
2. **Map Visuals to Ground Truth**: Once you identify a relevant visual card, locate its corresponding parent element in the **UI Element Data (JSON)**. The text *within* the card that makes it relevant must be present in the JSON data of that element or its children.
3. **Output the Card's Root Element (Minimal Enclosing Container)**: This is a critical rule. You must select the element that serves as the root for a single visual card. Think of it as the **smallest possible container** whose descendants (children, grandchildren, etc.) include **all** the text and UI elements for that one card, and nothing more. - **DO NOT** select a high-level container that groups multiple cards together (e.g., the entire list). - **DO NOT** select a specific element *inside* the card (e.g., just the title text or a button). - You **MUST** select the parent element that perfectly encapsulates one single, self-contained card.
4. **Comprehensive Selection**: If multiple distinct cards on the screen appear equally relevant to the subtask, you must select and output the root element 'name' for **every** one of them.
4 ˙Output Format
Your output must include the following two sections, strictly adhering to the format requirements:
Thought: Please provide a detailed analysis of your decision-making process. Clearly explain how you used the visual information from the screenshot to identify a relevant 'card'. Then, describe how you found the corresponding root element in the JSON and why it's the correct minimal container for that card, and why its contents are relevant to the 'Current Subtask'.
Output: Based on your analysis, output the filtered UI element names as a **JSON array (a list of strings)**. - The list should contain the 'name' (unique ID) of each selected card's root element.

A.3 Prompts for Talk Agent

Background
I will give you the user's instruction and a JSON object called 'categoryItems' which contains all relevant UI elements, grouped by category. Your task is to identify which category the user is asking about.
The user's instruction is: 'instruction'.
UI Categories and Items (categoryItems)
The UI information is provided in a 'categoryItems' JSON object. Its structure is as follows:
- The top-level keys are human-readable category names (e.g., "Restaurant Name", "Rating", "Monthly Sales"). **These are the exact category names you need to identify and output.** - The value for each category is a dictionary. Inside this dictionary, keys are element IDs, and values contain the element's display text. Here is the 'categoryItems' information for the current screen:
Your Task
1. Analyze the user's instruction. 2. Decide if the user wants to go back (Mode A) or get information about a category (Mode B). 3. If Mode B, determine which category name from the 'categoryItems' keys the user is referring to.
Mode Choosing
Analyze the user's intent to choose a mode.
- **A**: The user wants to go back to the previous step. - **B**: The user wants to know more information about one or more UI categories shown on the screen.
Displayed UI Choosing
If you chose 'B', which UI category is the user asking about? Your answer must be one of the top-level keys from the 'categoryItems' JSON provided above.
Output format
Your output consists of the following 2 parts:
Thought
Based on the requirements and information above, think about the behavior you are going to do. Output in English.
Mode
A or B
Displayed UI
If Mode is B, output the name of the category (or categories, comma-separated) that the user is interested in. This name **MUST** exactly match one of the top-level keys in the 'categoryItems' JSON. If Mode is A, leave this section empty.

Fig. 10. Prompts for Subtask Agent, Decision Agent and Talk Agent.

Table 5. Task performance and completion time across Baseline and FocalFlow conditions. Note: All numerical values represent completion time in minutes (min). ✓: Indicates a successful task completion. ×: Indicates a task failure. P8 and P9 were excluded from the paired t-test to ensure data consistency.

PID	Order†	Burger (Baseline)	Burger (FocalFlow)	Derma (Baseline)	Derma (FocalFlow)
P1	BL→FF, Derma→Burger	-	✓· 12.63	✓· 6.05	-
P2	BL→FF, Derma→Burger	-	✓· 9.34	✓· 13.73	-
P3	BL→FF, Derma→Burger	-	✓· 13.04	✓· 12.13	-
P4	FF→BL, Burger→Derma	-	✓· 12.24	× (Time Out) · 15.00	-
P5	BL→FF, Burger→Derma	✓· 14.51	-	-	✓· 14.96
P6	BL→FF, Burger→Derma	✓· 11.15	-	-	✓· 14.03
P7	BL→FF, Burger→Derma	✓· 12.24	-	-	✓· 14.04
P8	FF→BL, Burger→Derma	-	✓· 14.33	× (Early Quit) · 9.40	-
P9	FF→BL, Burger→Derma	-	× (Mismatch) · 5.93	× (Early Quit) · 7.41	-
P10	FF→BL, Derma→Burger	× (Time Out) · 15.00	-	-	✓· 11.07
P11	FF→BL, Derma→Burger	✓· 7.63	-	-	✓· 7.27
P12	FF→BL, Derma→Burger	✓· 11.86	-	-	✓· 14.31

† Counterbalancing order, e.g., BL→FF, Burger→Derma. BL = Baseline, FF = FocalFlow.